

Head and Neck

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Diseases of the head and neck range from the common cold to uncommon neoplasms of the ear and nose. The examples discussed in this chapter are organized based on the primary anatomic site at which they occur, that is, oral cavity, upper airways (including the nose, pharynx, larynx, and nasal sinuses), ears, neck, and salivary glands.

ORAL CAVITY

DISEASES OF TEETH AND SUPPORTING STRUCTURES

The oral cavity contains a complex microbial ecosystem that consists of more than 2000 bacterial, viral, and fungal species that are collectively termed the “oral microbiome.” Most of these microorganisms are

commensal in nature. However, the disruption of the composition, quantity, and function of this ecosystem may contribute to the development of local and systemic conditions/diseases. Oral microbiome dysbiosis has been associated with specific diseases of the oral cavity (dental caries, periodontitis, and oral cancer), which will be discussed later. Although there are some data suggesting an association between, oral dysbiosis and several systemic diseases including Alzheimer disease, low preterm birth weights, diabetes, cardiovascular disease, and cancers of other anatomic sites, including colorectal and pancreatic, neither the significance nor the mechanisms underlying these possible associations are defined.

Caries (Tooth Decay)

Dental caries is caused by focal demineralization of tooth structure (enamel and dentin) by acidic products of bacterial sugar fermentation.

Caries is the principal cause of tooth loss before age 35. Naturally, the development of dental caries can be affected by a number of factors, such as type of diet, bacteria, and other environmental agents. Caries was previously most common in high-resource countries with abundant carbohydrate-rich processed and refined foods. Demographics have, however, shifted due to both reduced incidence in high-resource countries and increased incidence in low-resource nations. The former reflects improved oral hygiene and, in the United States, drinking water fluoridation. Within the crystalline tooth enamel, fluoroapatite (calcium fluorophosphate) is more resistant to acid degradation than hydroxylapatite (calcium apatite). Increasing processed food consumption is associated with greater caries rates in low-resource countries, where the incidence is expected to continue to grow.

Gingivitis

Gingivitis is inflammation of the oral mucosa surrounding the teeth, caused by accumulation of dental plaque and calculus. It may occur at any age but is most prevalent and severe in adolescence. Dental plaque is a sticky, colorless biofilm that collects between and on the surface of the teeth because of poor oral hygiene. It contains a mixture of bacteria (which produce the acid that contributes to caries development), salivary proteins, and desquamated epithelial cells. If not removed, plaque can become mineralized to form calculus (tartar). Gingivitis is characterized by erythema, edema, bleeding, changes in contour of the teeth, and loss of soft tissue support for the teeth. Fortunately, it is reversible; therapy is primarily aimed at reducing the accumulation of plaque and calculus via regular oral hygiene.

Periodontitis

Periodontitis is an inflammatory process that affects the supporting structures of the teeth (periodontal ligaments), alveolar bone, and cementum. Sequelae of periodontitis include destruction of the periodontal ligament, which attaches teeth to the alveolar bone. This leads to loosening and eventual loss of teeth. Like caries, periodontitis is associated with poor oral hygiene but is also characterized by altered composition of the oral microbiome, which is believed to be important in pathogenesis. For the most part, facultative gram-positive organisms colonize healthy gingival sites, whereas plaque within areas of active periodontitis contains anaerobic and microaerophilic gram-negative flora. Of the 300 bacterial species within the oral cavity, adult periodontitis is associated primarily with *Aggregatibacter (Actinobacillus) actinomycetemcomitans*, *Porphyromonas gingivalis*, and *Prevotella intermedia*.

While localized to the gums, periodontitis is associated with a spectrum of systemic diseases, including acquired immunodeficiency syndrome (AIDS), Crohn disease, diabetes, Down syndrome, sarcoidosis, and disorders marked by neutrophil deficiency or dysfunction. Periodontal infections may also serve as a site of origin of bacteria that cause infective endocarditis and abscesses within the lungs and brain.

INFLAMMATORY/REACTIVE LESIONS

Aphthous Ulcers (Canker Sores)

Aphthous ulcers are common, often recurrent, and painful. The cause is not known. Oral aphthous ulcers affect up to 40% of the population and are most frequent in the first 2 decades of life. Their occurrence clusters within some families and they may be associated with immunologic disorders such as celiac disease, inflammatory bowel disease, and Behçet disease. The lesions appear as single or multiple, shallow, hyperemic, mucosal ulcerations covered by a thin exudate and rimmed by a narrow zone of erythema (Fig. 16.1). The underlying inflammatory infiltrate is initially mononuclear but



Fig. 16.1 Aphthous ulcer. Single ulceration with an erythematous halo surrounding a yellowish fibrinopurulent membrane.

becomes neutrophil-rich if there is secondary bacterial infection. Lesions typically resolve spontaneously in 7 to 10 days but sometimes persist for weeks, particularly in immunocompromised patients.

Fibrous Proliferative Lesions

Irritation Fibroma. The irritation fibroma, also called *traumatic fibroma* or *focal fibrous hyperplasia*, is a submucosal nodular mass of fibrous connective tissue that occurs primarily on the buccal mucosa along the bite line or the gingiva (Fig. 16.2). It is thought to be a reactive process induced by repetitive trauma. Treatment is complete surgical excision.

Pyogenic Granuloma. Pyogenic granuloma is typically found on the gingiva of children, young adults, and pregnant females (*pregnancy tumor*). This exophytic inflammatory lesion is red to purple in color and frequently ulcerated (Fig. 16.3). In some cases, growth can be so rapid to elicit concerns of malignancy. Histologically, pyogenic granulomas are a highly vascularized proliferation of organizing granulation tissue. Pyogenic granulomas can regress, mature into dense fibrous masses, or develop into a peripheral ossifying fibroma. Complete surgical excision is the definitive treatment.

Peripheral Ossifying Fibroma. Peripheral ossifying fibroma is a common gingival growth that is likely reactive in nature rather than neoplastic. As mentioned earlier, some arise from a long-standing pyogenic granuloma, and others develop de novo from cells of the periodontal ligament. Peripheral ossifying fibromas appear as red, ulcerated, nodular lesions of the gingiva. The highest incidence is in young females. Since the lesions have a recurrence rate of 15% to 20%, complete surgical excision down to the periosteum is required.



Fig. 16.2 Irritation fibroma. Smooth pink exophytic nodule on the buccal mucosa.



Fig. 16.3 Pyogenic granuloma. Erythematous and hemorrhagic exophytic mass arising from the gingival mucosa.

Peripheral Giant Cell Granuloma. Peripheral giant cell granuloma is an uncommon oral cavity lesion that likely represents a reactive inflammatory process. It is generally covered by intact gingival mucosa but may be ulcerated. Histologically, peripheral giant cell granulomas contain aggregates of multinucleate, foreign body–like giant cells separated by a fibroangiomatous stroma. Although not encapsulated, the lesions are usually well delimited and easily excised. They should be differentiated from central giant cell tumors found within the jaws and from the histologically similar but frequently multiple “brown tumors” seen in hyperparathyroidism (Chapter 24).

INFECTIONS

Herpes Simplex Virus Infections

Oral herpes usually presents as gingivostomatitis in children, pharyngitis in adults, and chronic mucocutaneous infection in immunocompromised individuals. Most orofacial herpetic infections are caused by herpes simplex virus type 1 (HSV-1), but oral HSV-2 (genital herpes) infections also occur. In children, primary infections are most common between 2 and 4 years of age. These are often asymptomatic but may present as acute herpetic gingivostomatitis, with abrupt onset of vesicles and ulcerations of the oral mucosa, in up to 20% of cases. These lesions can be accompanied by lymphadenopathy, fever, and anorexia. In adults, acute herpes pharyngitis is common and may recur.

MORPHOLOGY

Herpes vesicles range in size from a few millimeters to large bullae filled with clear, serous fluid. They rapidly rupture to become painful, red-rimmed, shallow ulcerations. Intracellular and intercellular edema and acantholysis create clefts that may become macroscopic vesicles. Individual epidermal cells at vesicle margins or free within the fluid may contain eosinophilic **intranuclear viral inclusions** (see Fig. 8.12), or several cells may fuse to produce giant cells (**multinucleate polykaryons**). These can be demonstrated by the diagnostic **Tzanck test** based on microscopic examination of the vesicle fluid. Vesicles and shallow ulcers usually clear spontaneously within 3 to 4 weeks, but the virus treks along regional nerves and eventually becomes dormant (i.e., latent) within local ganglia (e.g., the trigeminal ganglion).

Most individuals who have been infected harbor latent virus in epithelial cells or ganglia. Reactivation of latent HSV causes *recurrent herpetic stomatitis* and the triggers for reactivation include trauma, allergies, exposure to ultraviolet light, upper respiratory tract infection, pregnancy, menstruation, immunosuppression, and exposure to temperature extremes. In contrast to acute herpetic gingivostomatitis,

recurrent herpetic stomatitis occurs at the site of primary inoculation or in mucosa associated with the same ganglion. The lesions appear as groups of small (1 to 3 mm) vesicles on the lips (*herpes labialis*), nasal orifices, buccal mucosa, gingiva, and hard palate. Although they typically resolve in 7 to 10 days, lesions can persist in immunocompromised patients and may require systemic antiviral therapy. Recurrent herpetic infection can also cause corneal ulcerations (Chapter 8).

Other viral infections that can involve the oral cavity as well as the head and neck region include herpes zoster, Epstein-Barr virus (EBV; infectious mononucleosis, nasopharyngeal carcinoma, lymphoma), cytomegalovirus, enterovirus (herpangina, hand-foot-and-mouth disease, acute lymphonodular pharyngitis), and rubeola (measles).

Oral Candidiasis (Thrush)

Candida albicans is a normal component of the oral flora in approximately 50% of the population and is the most common fungal infection of the oral cavity. Factors that influence the likelihood of infection include the strain of *C. albicans*, oral microbiome composition, and the individual's immune status. Oral candidiasis can be *pseudomembranous*, *erythematous*, or *hyperplastic*. The pseudomembranous form, *thrush*, is characterized by a superficial, gray to white inflammatory membrane composed of matted organisms enmeshed in a fibrinosuppurative exudate that can be readily scraped off to reveal an underlying erythematous inflammatory base. The infection typically remains superficial except in the setting of immunosuppression, such as in individuals with organ or hematopoietic stem cell transplants, neutropenia, chemotherapy-induced immunosuppression, AIDS, or diabetes. Broad-spectrum antibiotics that eliminate or alter the normal bacterial flora of the mouth can also promote thrush.

Deep Fungal Infections

In addition to their usual sites of infection, some deep fungal infections, including histoplasmosis, blastomycosis, coccidioidomycosis, cryptococcosis, zygomycosis, and aspergillosis, have a predilection for the oral cavity, head, and neck. The incidence of oral fungal infections has grown along with increasing numbers of immunocompromised patients because of diseases such as AIDS, therapies for cancer, and organ transplantation.

ORAL MANIFESTATIONS OF SYSTEMIC DISEASE

Oral lesions are often the first sign of underlying systemic conditions. Some of the more common disease associations and their oral manifestations are cited in Table 16.1. Only hairy leukoplakia is considered in more detail here.

Hairy Leukoplakia

Hairy leukoplakia is a distinctive oral lesion on the tongue caused by EBV that usually occurs in immunocompromised patients. In patients infected with the human immunodeficiency virus (HIV), hairy leukoplakia may portend development of AIDS. However, it is increasingly observed in patients who are immunocompromised for other reasons, including cancer therapy, transplant-associated immunosuppression, and advanced age. Hairy leukoplakia takes the form of white, confluent patches of fluffy (“hairy”), hyperkeratotic thickenings, almost always situated on the lateral border of the tongue. Unlike thrush, the lesion cannot be scraped off. The distinctive microscopic appearance consists of hyperparakeratosis and acanthosis with “balloon cells” in the upper spinous layer. EBV RNA transcripts and proteins can be detected within the lesional cells. Superimposed candidal infection can add to the “hairiness.”

TABLE 16.1 Oral Manifestations of Some Systemic Diseases

Systemic Disease	Associated Oral Changes
Infectious Diseases	
Scarlet fever	Fiery red tongue with prominent papillae (raspberry tongue); white-coated tongue through which hyperemic papillae project (strawberry tongue)
Measles	Spotty enanthema in the oral cavity often precedes the skin rash; ulcerations on the buccal mucosa about Stensen duct produce Koplik spots
Infectious mononucleosis	Acute pharyngitis and tonsillitis coated with a gray-white exudative membrane; enlargement of lymph nodes in the neck; palatal petechiae
Diphtheria	Characteristic dirty white, fibrinosuppurative, tough, inflammatory membrane over the tonsils and retropharynx
Human immunodeficiency virus	Predisposition to opportunistic oral infections, particularly herpes virus, Candida, and other fungi; oral lesions of Kaposi sarcoma and hairy leukoplakia (see text)
Dermatologic Conditions^a	
Lichen planus	Reticulate, lacelike, white keratotic lesions that sometimes ulcerate and rarely form bullae; seen in more than 50% of patients with cutaneous lichen planus; rarely is the sole manifestation
Pemphigus	Vesicles and bullae prone to rupture, leaving hyperemic erosions covered with exudates
Bullous pemphigoid	Oral lesions (mucus membrane pemphigoid) resemble those of pemphigus but can be differentiated histologically
Erythema multiforme	Maculopapular, vesiculobullous eruption that sometimes follows an infection elsewhere, ingestion of drugs, development of cancer, or a collagen vascular disease; when there is widespread mucosal and skin involvement, referred to as Stevens-Johnson syndrome
Hematologic Disorders	
Pancytopenia (agranulocytosis, aplastic anemia)	Severe oral infections in the form of gingivitis, pharyngitis, tonsillitis; may extend to produce cellulitis of the neck (Ludwig angina)
Acute leukemia	With depletion of functioning neutrophils, oral lesions similar to those in pancytopenia
Monocytic leukemia	Leukemic infiltration and enlargement of the gingivae, often with accompanying periodontitis
Miscellaneous	
Melanotic pigmentation	May appear in Addison disease, hemochromatosis, fibrous dysplasia of bone (Albright syndrome), and Peutz-Jeghers syndrome (gastrointestinal polyposis)
Phenytoin (Dilantin) ingestion	Striking fibrous enlargement of the gingivae
Pregnancy	Pyogenic granuloma protruding from the gingiva (pregnancy tumor)
Rendu-Osler-Weber syndrome	Autosomal dominant disorder with multiple congenital aneurysmal telangiectasias beneath mucosal surfaces of the oral cavity and lips

^aSee Chapter 25.

PRECANCEROUS AND CANCEROUS LESIONS

Many epithelial and connective tissue tumors of the head and neck region (e.g., papillomas, hemangiomas, lymphomas) occur elsewhere in the body and are described in other chapters. This discussion considers only the most common oral cancer, squamous cell carcinoma (SCC), and its associated precancerous lesions.

Leukoplakia and Erythroplakia

Leukoplakia is defined by the World Health Organization as “a white patch or plaque that cannot be scraped off and cannot be characterized clinically or pathologically as any other disease.” This clinical term is reserved for lesions that are present in the oral cavity for no apparent reason. As such, white patches caused by obvious irritation or entities such as lichen planus and candidiasis are not considered to be leukoplakia. Approximately 3% of the world’s population have leukoplakia; 5% to 25% of these lesions are premalignant. Thus, until proven otherwise by histologic evaluation, all leukoplakias must be considered precancerous.

Related to leukoplakia, but much less common and more ominous, is *erythroplakia*, a red, velvety, possibly eroded area within the oral

cavity that usually remains level with or may be slightly depressed relative to the surrounding mucosa (Fig. 16.4). The epithelium in such lesions tends to be markedly atypical, and the risk of malignant transformation is much higher than with leukoplakia. Intermediate forms that have the characteristics of both leukoplakia and erythroplakia are termed *speckled erythroplakia*.

Both leukoplakia and erythroplakia may be seen in adults at any age, but they are usually found in persons 40 to 70 years of age, with a 2:1 male preponderance. Although these lesions have multifactorial origins, the use of tobacco (cigarettes, pipes, cigars, and certain forms of smokeless tobacco) is a common antecedent.

MORPHOLOGY

Leukoplakia may occur anywhere in the oral cavity (favored locations are buccal mucosa, floor of the mouth, ventral surface of the tongue, palate, and gingiva). The lesions appear as solitary or multiple white patches or plaques, often with sharply demarcated borders. They may be slightly thickened and smooth, wrinkled and fissured, or may appear as raised, sometimes corrugated, verrucous plaques (Fig. 16.5A). On histologic examination, they present

a spectrum of epithelial changes ranging from hyperkeratosis overlying a thickened, acanthotic but orderly mucosal epithelium to lesions with markedly dysplastic changes, sometimes merging into carcinoma in situ (Fig. 16.5B).

The histologic changes in **erythroplakia** only rarely demonstrate orderly epidermal maturation; virtually all (approximately 90%) display severe dysplasia, carcinoma in situ, or minimally invasive carcinoma. An intense subepithelial inflammatory reaction with vascular dilation that contributes to the reddish appearance of these lesions is often present.

Squamous Cell Carcinoma

Approximately 95% of cancers of the head and neck are SCCs, with the remainder largely consisting of adenocarcinomas of salivary gland origin. Head and neck squamous cell carcinoma (HNSCC) is the sixth most common neoplasm in the world. At current rates, approximately 50,000 cases in the United States and more than 650,000 cases worldwide are diagnosed each year. The term *head and neck cancer* includes tumors of the oral cavity, pharynx, larynx, and nasal cavities, which are discussed here, and also tumors of the thyroid (Chapter 24) and salivary glands (discussed later in this chapter).

The pathogenesis of SCC is multifactorial.

- *Infection with high-risk human papillomavirus (HPV)* is now the primary cause of SCC of the oropharynx.
- Within North America and Europe, oral cavity SCC has classically been a disease of middle-aged individuals who have been chronic users of *smoked tobacco* and *alcohol*.



Fig. 16.4 Erythroplakia. (A) Red discoloration of the maxillary gingiva. (B) Red lesion of the mandibular alveolar ridge. Biopsy of both lesions revealed carcinoma in situ.

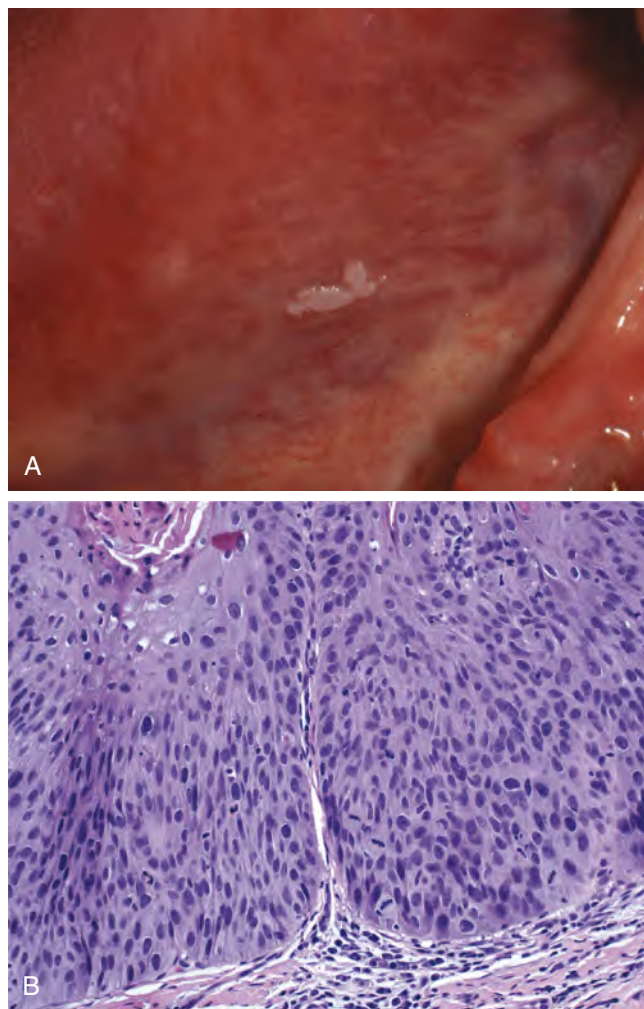


Fig. 16.5 Leukoplakia. (A) Clinical appearance of leukoplakia is highly variable. In this example, the lesion is relatively smooth and thin with well-demarcated borders. (B) Histologic appearance of leukoplakia showing severe dysplasia that is characterized by nuclear and cellular pleomorphism, numerous mitotic figures, and a loss of normal maturation.

- In India and Asia, the chewing of betel quid and paan is a major regional predisposing influence. This concoction, considered a delicacy by some, contains ingredients such as areca nut, slaked lime, and tobacco, wrapped in a betel leaf; many of the ingredients of paan can give rise to carcinogens.
- *Actinic radiation* (sunlight) and pipe smoking are known predisposing influences for cancer of the lower lip.
- The incidence of oral cavity SCC (particularly involving the tongue) in individuals younger than 40 years of age, who have no known risk factors, has been on the rise. The pathogenesis in this group of patients, who are nonsmokers and are not infected with HPV, is unknown.

In the oropharynx, as many as 90% of SCCs, particularly those involving the tonsils, the base of the tongue, and the pharynx, harbor oncogenic variants of HPV, particularly HPV-16. HPV-associated SCC of the oropharynx has increased more than two-fold over the past 2 decades. It is hoped that this trend will be reversed by the HPV vaccine, which is protective against HPV-associated cervical cancer. Early detection of HPV-associated head and neck SCC can be particularly

challenging because the anatomic sites of origin (tonsillar crypts, base of tongue, and oropharynx) are not readily accessible or amenable to cytologic screening (unlike the cervix) for premalignant lesions. Unlike the oropharynx, HPV-associated SCC of the oral cavity is relatively uncommon.

Prognosis is dependent on several factors including the specific etiology of SCC. The 5-year survival rate of “classic” smoking- and alcohol-related early-stage SCC is approximately 80%, but survival drops to 20% for late-stage disease. Long-term survival is better in patients with HPV-positive SCC. The dismal outlook in classic SCC reflects diagnosis at advanced stage as well as the frequent presence of multiple primary tumors. Notably, new primary tumors occur at 3% to 7% per year, the highest rate of among all malignancies. This observation has led to the concept of *field cancerization*, which postulates that multiple individual primary tumors develop independently in the upper aerodigestive tract because of years of mucosal exposure to carcinogens. This may explain why there is an almost 35% chance of developing at least one new primary tumor in individuals who survive 5 years after detection of the initial primary tumor. Development of these tumors can be particularly devastating for individuals whose initial lesions were small. The second primary tumors are commonly fatal. Early detection of premalignant lesions is therefore critical for long-term survival of these patients.

Molecular Biology of Squamous Cell Carcinoma. As with other cancers, SCC development is driven by the accumulation of mutations and epigenetic changes that alter the expression and function of oncogenes and tumor suppressor genes. Large-scale exome-sequencing studies, including those performed as part of The Cancer Genome Atlas (TCGA), have defined the mutational landscape of SCC. Several important findings have come from these studies. First, some genes previously proposed as playing critical roles in SCC (*TP53*, *CDKN2A*, *PIK3CA*) are mutated with frequencies sufficient to suggest that they are drivers of cancer development. Second, a number of novel genetic alterations have been identified, particularly loss of function mutations in *NOTCH1* and downstream signaling components and the tumor suppressor *FAT1*, a member of the cadherin family. Third, although *TP53* is the most frequently mutated gene, the mutational frequency for the next most common genes ranged from 1% to 23%, indicating considerable intertumor variability with respect to specific driver mutations. Finally, the mutations identified in HPV-positive and HPV-negative tumors are different. For example, HPV-negative tumors harbor more somatic mutations than HPV-positive tumors. In addition, owing to the expression of the HPV oncoproteins E6 and E7, *TP53* and *RB* mutations are typically absent in HPV-positive tumors, similar to what is observed in HPV-positive cervical cancer (Chapter 22). HPV-positive oropharyngeal SCCs are generally associated with better outcomes despite frequent local lymph node metastases.

Lesions are considered premalignant and at risk for progressing to HNSCC when histologic dysplasia is present. However, the criteria for diagnosing dysplasia remain subjective and open to a wide range of interpretation, even among highly qualified pathologists. This, in part, reflects the absence of validated histologic findings that predict malignant transformation of dysplastic lesions. In general, ~15% of premalignant lesions will progress to HNSCC with a mean time interval of 5 years. Our inability to accurately prognosticate based on histologic alterations emphasizes the need to develop molecular criteria for predicting outcomes. HNSCCs are treated with surgery and postoperative radiation therapy. Novel combination of immunotherapies including checkpoint inhibitors have shown promising results in HNSCC.

MORPHOLOGY

SCC may arise anywhere in the head and neck region that is lined by stratified squamous epithelium. For “classic” HPV-negative keratinizing SCC, the favored locations are the ventral surface of the tongue, floor of the mouth, lower lip, soft palate, and gingiva (Table 16.2). By contrast, HPV-associated SCCs are most often nonkeratinizing neoplasms arising in the epithelium of the tonsillar crypts within the lingual tonsils, base of tongue, soft palate, and pharynx. Although keratinizing SCCs are typically preceded by premalignant lesions, such as leukoplakia and erythroplakia, HPV-associated SCCs tend to develop without a readily identified premalignant (i.e., dysplastic) component.

Early-stage keratinizing SCCs appear as raised, firm, pearly plaques or irregular, roughened, or verrucous areas of mucosal thickening. Either pattern may be superimposed on a background of leukoplakia or erythroplakia. As these lesions enlarge, they typically form ulcerated and protruding masses with irregular and indurated, or rolled, borders (Fig. 16.6A). By contrast, HPV-associated SCCs typically present as small primary tumors that lack obvious surface mucosal involvement and instead are accompanied by cervical lymphadenopathy due to metastatic spread.

Keratinizing SCCs begin as dysplastic lesions, which may or may not progress to full-thickness dysplasia (carcinoma in situ) before invading the underlying connective tissue stroma (Fig. 16.6B). This difference in progression contrasts with cervical cancer (Chapter 22), in which full-thickness dysplasia, representing carcinoma in situ, typically develops before invasion. Keratinizing SCCs range from well-differentiated to anaplastic, sometimes sarcomatoid, tumors. However, the degree of histologic differentiation, as determined by the relative degree of keratinization, is not correlated with behavior, including growth rate. SCC tends to infiltrate locally before metastasizing. The routes of extension depend on the primary site. The cervical lymph nodes are favored sites of local metastasis, and the most common sites of distant metastasis are mediastinal lymph nodes, lungs, liver, and bones. Unfortunately, distant metastases are often present at the time of discovery of the primary lesion.

The histology of **HPV-associated SCC is characterized by the proliferation of nests and lobules of nonkeratinizing and basaloid cells** growing within sheets of lymphocytes (Fig. 16.7A). Immunohistochemical detection of high-level expression of p16, a cyclin-dependent kinase inhibitor, is strongly correlated with the presence of HPV and can serve as a marker for HPV-associated SCC (Fig. 16.7B). However, confirmatory testing, using PCR or in situ hybridization (ISH) (Fig. 16.7C), may be required in certain cases.

TABLE 16.2 Differences Between HPV-Associated and Non-HPV-Associated SCC

	HPV-Associated SCC	Non-HPV-Associated SCC
Patient age	Younger	Older
Risk factors	Number of oral sex partners	Tobacco, alcohol
Location	Oropharynx	Oral cavity
Clinical presentation	Small primary lesion with bulky nodal disease	Large primary lesion with variable nodal disease
Histology	Nonkeratinizing SCC	Keratinizing SCC
Distant metastasis	Rare	Common
Clinical outcomes	Good	Poor
Risk of second primary	Low	High

HPV, Human papillomavirus; SCC, squamous cell carcinoma.

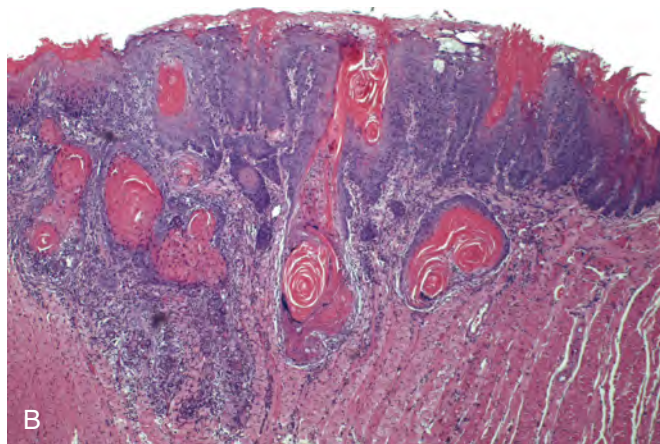


Fig. 16.6 Keratinizing squamous cell carcinoma. (A) Clinical appearance demonstrating ulceration and induration of the oral mucosa. (B) Histologic appearance demonstrating numerous nests and islands of malignant keratinocytes with keratin whorls that invade the underlying connective tissue stroma and skeletal muscle.

ODONTOGENIC CYSTS AND TUMORS

The overwhelming majority of odontogenic cysts are derived from remnants of odontogenic epithelium present within the jaws. In contrast to the rest of the skeleton, epithelial-lined cysts are common in the jaws. In general, they are categorized as being found in three distinct locations (Fig. 16.8): (1) associated with the crown of an unerupted tooth, (2) associated with the apex of a tooth, and (3) not specifically associated with a tooth. These cysts are subclassified as either inflammatory or developmental (Table 16.3), with only the most common described here.

The *dentigerous cyst* originates around the crown of an unerupted tooth because of fluid accumulation between the developing tooth and the dental follicle. Radiographically, these are unilocular lesions most often associated with impacted third molar (wisdom) teeth. Histologically, they are lined by a thin layer of stratified squamous epithelium, and there is frequently a dense chronic inflammatory cell infiltrate in surrounding connective tissue. Complete surgical excision is curative.

The *odontogenic keratocyst* must be differentiated from other odontogenic cysts because of its aggressive behavior. Odontogenic keratocysts can be seen at any age but are most common between 10 and 40 years of age, in males, and can be observed in any region of the maxilla or mandible. The lesions are well-defined unilocular or multilocular radiolucencies with a lining consisting of a thin layer of keratinized stratified squamous epithelium, a prominent basal cell layer, and a corrugated epithelial surface. Treatment requires complete excision

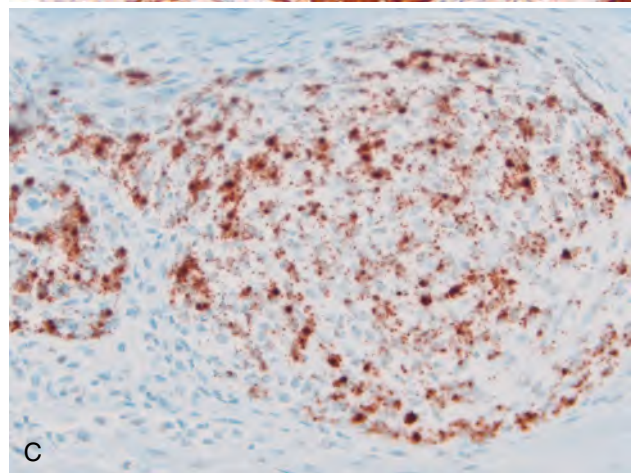
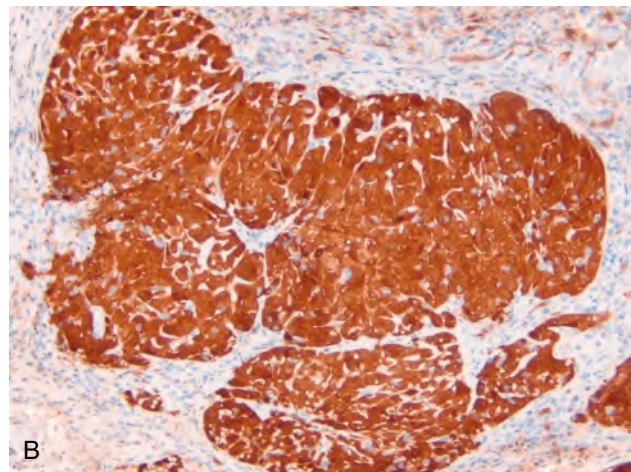
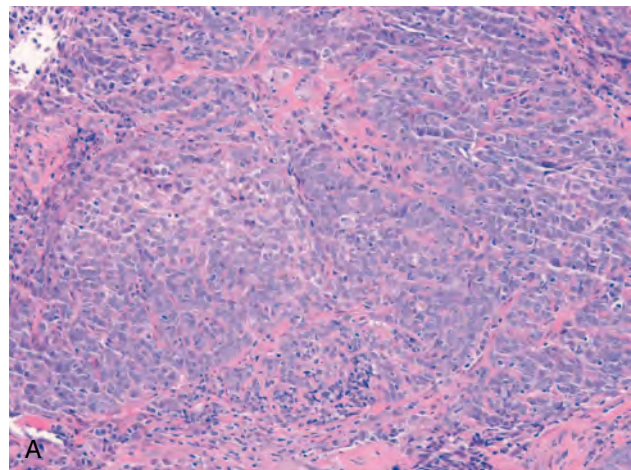


Fig. 16.7 Nonkeratinizing squamous cell carcinoma. (A) Histologic appearance demonstrating nests and cords of basaloid appearing cells. (B) Immunohistochemistry demonstrating strong nuclear and cytoplasmic tumor cell staining for p16. (C) In situ hybridization for high-risk human papillomavirus mRNA expression.

because of locally aggressive behavior. Recurrence rates for inadequately removed lesions can be as high as 60%. About 80% of the lesions are solitary, but patients with multiple cysts should be evaluated for nevoid basal cell carcinoma syndrome (Gorlin syndrome), which is caused by germline loss of function mutations in the tumor suppressor gene *PTCH* (Patched) (Chapter 25).

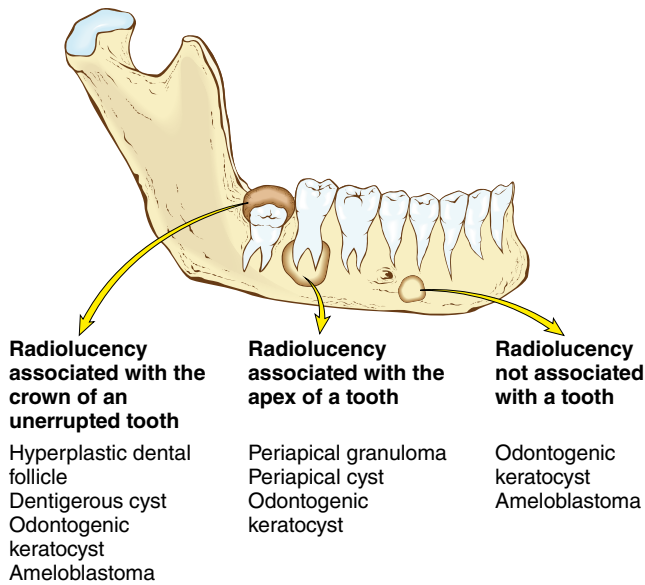


Fig. 16.8 Odontogenic cysts. Diagrammatic comparison of the anatomic location of common inflammatory and developmental odontogenic cysts.

TABLE 16.3 Histologic Classification of Odontogenic Cysts

Inflammatory
Radicular cyst
Paradental (inflammatory collateral) cyst
Developmental
Dentigerous cyst
Odontogenic keratocyst
Gingival cyst
Lateral periodontal cyst
Glandular odontogenic cyst
Calcifying odontogenic cyst
Orthokeratinized odontogenic cyst

The *radicular cyst* (or *periapical cyst*) is a common inflammatory lesion found at the tooth apex. The cysts develop because of long-standing inflammation of the tooth (*pulpitis*), often secondary to advanced carious lesions or local trauma. Necrosis of the pulp tissue may occur and can traverse the length of the root to exit at the apex into the surrounding alveolar bone. Over time, granulation tissue may develop, with subsequent epithelialization that results in a radicular cyst. The older term *periapical granuloma* is still used, even though the lesion does not show true granulomatous inflammation. Periapical inflammatory lesions persist due to continued presence of bacteria or other irritating agents. Successful treatment, therefore, necessitates the complete removal of offending material and restoration or extraction of the tooth.

Odontogenic tumors are a group of lesions with diverse histologic appearances and clinical behavior. Some are true neoplasms (both benign and malignant), and others are more likely hamartomas. Odontogenic tumors are derived from odontogenic epithelium, odontogenic mesenchyme, or both (Table 16.4). The two most common and clinically significant tumors are odontoma and ameloblastoma.

- *Odontoma*, the most common odontogenic tumor, arises from epithelium and is associated with extensive enamel and dentin

TABLE 16.4 Histologic Classification of Odontogenic Tumors

Tumors of Odontogenic Epithelium
Benign
Adenomatoid odontogenic tumor
Squamous odontogenic tumor
Calcifying epithelial odontogenic tumor
Ameloblastoma: conventional, unicystic, and extrasosseous
Adenoid ameloblastoma
Malignant
Ameloblastic carcinoma
Sclerosing odontogenic carcinoma
Clear-cell odontogenic carcinoma
Ghost cell odontogenic carcinoma
Primary intraosseous squamous cell carcinoma
Tumors of Odontogenic Mesenchyme (Benign)
Odontogenic fibroma
Odontogenic myxoma
Cementoblastoma
Cemento-ossifying fibroma
Tumors of Odontogenic Epithelium and Mesenchyme
Benign
Odontoma, compound type
Odontoma, complex type
Ameloblastic fibroma
Ameloblastic fibro-odontoma
Dentinogenic ghost cell tumor
Malignant
Ameloblastic fibrosarcoma
Odontogenic carcinosarcoma

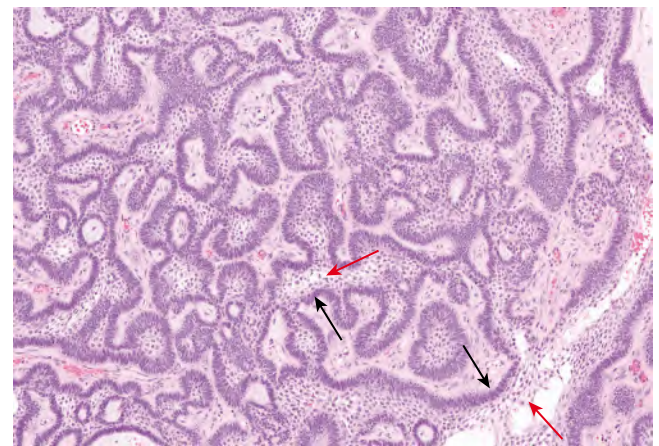


Fig. 16.9 Ameloblastoma is composed of odontogenic epithelium demonstrating peripheral columnar cells (*black arrows*) with hyperchromatic nuclei exhibiting reverse polarization with central areas of loosely arranged suprabasal cells resembling the stellate reticulum (*red arrows*) of the enamel organ.

deposition. These lesions probably represent hamartomas rather than true neoplasms and are cured by local excision.

- *Ameloblastoma* (Fig. 16.9) arises from odontogenic epithelium and does not display ectomesenchymal differentiation. (Ectomesenchymal cells, derived from the neural crest, give rise to major dental structures.) It is cystic, slow growing, and locally invasive with a typically indolent course. Wide surgical resection is required to prevent recurrence.

KEY CONCEPTS

Oral Cavity

- Caries is the most common cause of tooth loss in persons younger than 35 years of age. The primary cause is tooth destruction by acid end-products of bacterial sugar fermentation.
- Gingivitis is a common, reversible inflammation of the mucosa surrounding the teeth.
- Periodontitis is a chronic inflammatory condition that results in destruction of the supporting structures and eventual tooth loss. It is associated with poor oral hygiene and altered oral microbiota.
- Aphthous ulcers are painful superficial ulcers of unknown etiology.
- Fibromas and pyogenic granulomas are common reactive lesions of the oral mucosa.
- Oral candidiasis is generally associated with immune suppression.
- Leukoplakias and erythroplakias are oral mucosal lesions that may undergo malignant transformation.
- The majority of oral cavity and oropharyngeal cancers are SCCs. The two major risk factors are tobacco and alcohol use and high-risk HPV infection.

UPPER AIRWAYS

The term *upper airway* is used here to include the nose, pharynx, and larynx. Disorders of these structures are among the most common afflictions of humans, but fortunately the overwhelming majority are more nuisance than threat.

NOSE

Inflammatory diseases, such as the common cold, are the most frequent disorders of the nose and accessory air sinuses. These are most often viral in origin but can be complicated by superimposed bacterial infections. Destructive inflammatory nasal diseases and primary tumors of the nasal cavity or maxillary sinus occur less frequently.

Inflammatory Lesions

Infectious Rhinitis

Infectious rhinitis, also known as the *common cold*, is caused by several viruses. Major offenders are adenoviruses, echoviruses, and rhinoviruses, all of which evoke a profuse catarrhal discharge. During the initial acute stages, the nasal mucosa is thickened, edematous, and red, the nasal cavities are narrowed, and the turbinates are enlarged. These

changes may spread to produce pharyngotonsillitis. Secondary bacterial infection enhances the inflammatory reaction and produces a mucopurulent or sometimes suppurative exudate. Fortunately, these infections clear rapidly. As the saying goes, “in a week if treated, but in 7 days if ignored.” In recent years, an important differential diagnosis with the common cold has become a SARS-CoV-2 infection (COVID-19). Upper respiratory symptoms of COVID-19 infection overlap with those of the common cold but are often accompanied by taste and smell disturbances, lower respiratory manifestations (cough), and fever.

Allergic Rhinitis

Allergic rhinitis (hay fever) is initiated by hypersensitivity to one of a large group of allergens, most commonly plant pollens, fungi, animal allergens, and dust mites. It affects 20% of the US population. As with asthma, allergic rhinitis is an IgE-mediated immune reaction with an early- and late-phase response (see “Immediate [Type I] Hypersensitivity” in Chapter 6). The allergic reaction is characterized by mucosal edema, erythema, and mucus secretion accompanied by eosinophil-rich leukocytic infiltrates.

Nasal Polyps

Recurrent attacks of rhinitis may eventually lead to focal protrusions of the mucosa, producing *nasal polyps*, which may reach 3 to 4 cm in length. Histologically, these polyps consist of edematous mucosa with loose stroma, hyperplastic or cystic mucus glands, and infiltrates of neutrophils, eosinophils, and plasma cells (Fig. 16.10). In the absence of bacterial infection, the mucosal surface is intact, but it may become ulcerated. When multiple or large, nasal polyps may encroach on the airway and impair sinus drainage. Despite features indicating an allergic etiology, most people with nasal polyps are not atopic, and only 0.5% of atopic patients develop nasal polyps.

Chronic Rhinitis

Chronic rhinitis follows repeated episodes of microbial or allergic rhinitis with the eventual development of superimposed bacterial infection. A deviated nasal septum or nasal polyps that impair drainage of secretions contribute to the likelihood of microbial invasion. Superficial desquamation or ulceration of the mucosal epithelium is common. Inflammatory infiltrates include variable numbers of neutrophils, lymphocytes, and plasma cells. The infection may extend into the sinuses.

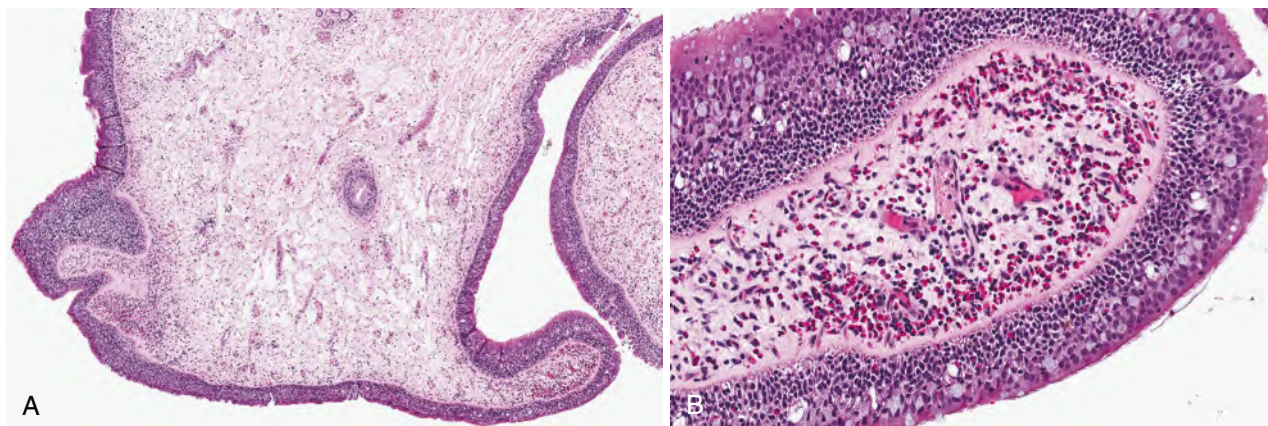


Fig. 16.10 (A) Nasal polyps. Low-power magnification showing edematous stroma lined by epithelium. (B) High-power view showing stromal edema, eosinophil-rich inflammatory infiltrate, and respiratory epithelial lining.

Sinusitis

Most cases of acute sinusitis are preceded by rhinitis, but maxillary sinusitis occasionally arises by extension of a periapical infection through the bony sinus floor. Responsible microorganisms are usually components of the oral microbes that elicit an inflammatory reaction. The resulting mucosal edema impairs sinus drainage and may result in *empyema* of the sinus. Outflow obstruction occurs most often in the frontal and, less commonly, anterior ethmoid sinuses. This may occasionally lead to mucus accumulation, producing a so-called *mucocoele*.

Acute sinusitis may progress to *chronic sinusitis*, particularly when drainage is disrupted. A polymicrobial infection, largely composed of normal oral microbial components, is present in most cases. Fungi may cause severe chronic sinusitis (e.g., in mucormycosis), especially in diabetic patients. Sinusitis can, uncommonly, present as a component of *Kartagener syndrome*, which also includes bronchiectasis and situs inversus (Chapter 15), due to inherited genetic defects in ciliary action. Although most cases of chronic sinusitis are more uncomfortable than disabling or serious, the infections can occasionally spread into the orbit or penetrate surrounding bone, leading to osteomyelitis. Further spread into the cranial vault can cause septic thrombophlebitis within a dural venous sinus.

Necrotizing Lesions of the Nose and Upper Airways

Necrotizing ulcerating lesions of the nose and upper respiratory tract may be produced by:

- *Acute invasive fungal infections* (including mucormycosis; Chapter 8), particularly in diabetic and immunosuppressed patients
- *Granulomatosis with polyangiitis*, previously called Wegener granulomatosis (Chapter 11)
- *Extranodal NK/T-cell lymphoma, nasal type*, in which the tumor cells harbor EBV (Chapter 13). This neoplasm typically occurs in males of Asian or Latin American descent in the fifth or sixth decade of life and is frequently complicated by ulceration and superimposed bacterial infection. Untreated, these neoplasms are almost always rapidly fatal because of local tumor spread and cranial vault penetration or secondary bacterial infection with septic dissemination. Localized cases can often be controlled with radiotherapy, but relapse and recurrence are common and associated with poor outcomes.

NASOPHARYNX

The nasopharyngeal mucosa and related lymphoid structures and glands may be involved in a wide variety of infections (e.g., diphtheria, infectious mononucleosis) as well as neoplasms.

Inflammatory Lesions

Pharyngitis and tonsillitis are frequently caused by viral upper respiratory infections. Rhinoviruses, echoviruses, and adenoviruses are the most common causes; remaining cases are primarily due to various strains of influenza or respiratory syncytial virus. Erythema and edema of the nasopharyngeal mucosa with reactive enlargement of nearby tonsils and lymph nodes are characteristic. Bacterial infection may cause pharyngitis and tonsillitis or, alternatively, occur as a secondary process superimposed on viral infection. β -hemolytic streptococci are the most frequent pathogens, but *Staphylococcus aureus* or other bacteria may also be involved. When bacteria are culpable, exudate and exudative membranes (pseudomembranes) may cover the nasopharyngeal mucosa as well as enlarged nasopalatine and palatine tonsils. *Follicular tonsillitis*, in which enlarged tonsils (due to reactive lymphoid hyperplasia) are dotted by exudate emanating from the tonsillar crypts, is common. Whether recurrent episodes of acute

tonsillitis favor the development of chronic tonsillar enlargement is open to debate, but regardless of the cause of the enlargement, patients may benefit from surgical excision.

The major importance of streptococcal “sore throat” lies in the possible development of late sequelae, such as rheumatic fever (Chapter 12) and glomerulonephritis (Chapter 20).

TUMORS OF THE NOSE, SINUSES, AND NASOPHARYNX

Tumors in the nose, sinuses, and nasopharynx are infrequent but include a wide spectrum of mesenchymal and epithelial neoplasms.

Sinonasal Angiofibroma

Sinonasal angiofibroma is a richly vascular tumor that occurs almost exclusively in adolescent males who often present with epistaxis. It is believed to arise within the fibrovascular stroma of the posterolateral wall and roof of the nasal cavity. It consists of numerous vessels embedded in a dense collagenous stroma (Fig. 16.11). Surgical removal, often with preoperative embolization to decrease bleeding, is the treatment of choice. Although benign, nasopharyngeal angiofibroma is often locally aggressive, sometimes extending into the cranium. Recurrence rates can approach 20%, and 9% of cases can be fatal. Mutations in *CTNFB1*, which encodes β -catenin, are present in the majority of sporadic nasopharyngeal angiofibromas. This tumor can also develop or syndromically in association with familial adenomatous polyposis (in such cases the tumor is driven by germline *APC* gene mutations, which also lead to β -catenin activation).

Sinonasal Papilloma

Sinonasal papilloma is a benign neoplasm arising from the respiratory or Schneiderian mucosa lining the nasal cavity and paranasal sinuses. These lesions occur in three forms: *inverted* (or *endophytic*, the most common variant), *oncocytic*, and *exophytic*. Sinonasal papillomas are most common in males between 30 and 60 years of age. The inverted form shows downward, bulbous, pushing growth into the underlying stroma (Fig. 16.12). Although this is a benign neoplasm, it can display locally aggressive behavior within both the nose and the paranasal sinuses, including invasion into the orbit or cranial vault, and it has a high rate of recurrence if not adequately excised. Malignant transformation occurs in up to 5% of cases. Most inverted sinonasal papillomas have *EGFR* gene mutations. The remaining cases generally harbor HPV DNA, often low-risk types 6 and 11.

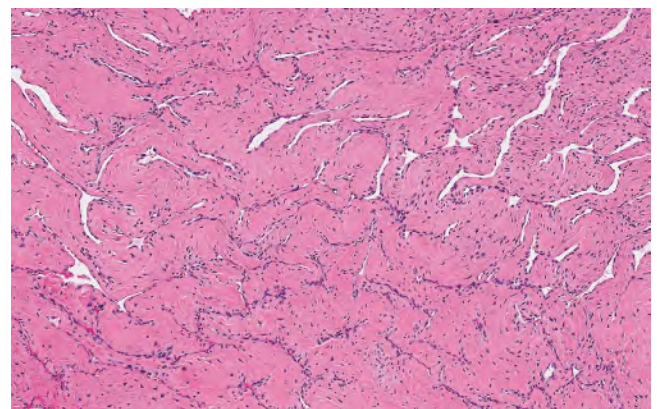


Fig. 16.11 Sinonasal angiofibroma showing thin-walled vessels, lined by a single layer of endothelial cells, and separated by dense collagenous stroma containing small, bland fibroblasts.

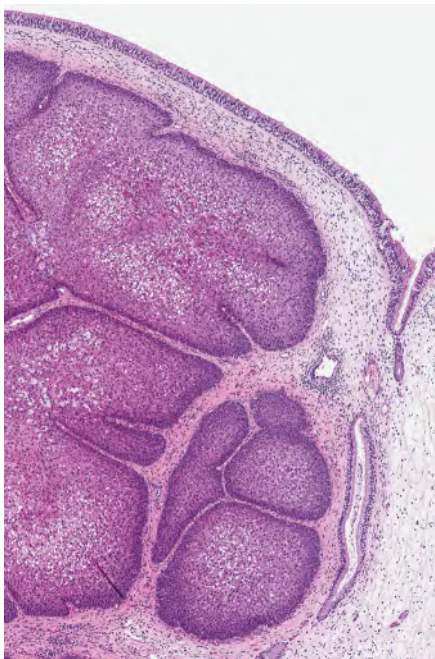


Fig. 16.12 Sinonasal papilloma (inverted papilloma). The overlying surface is lined by thin respiratory epithelium; however, the underlying stroma contains multiple nodules of thick neoplastic epithelium growing inward.

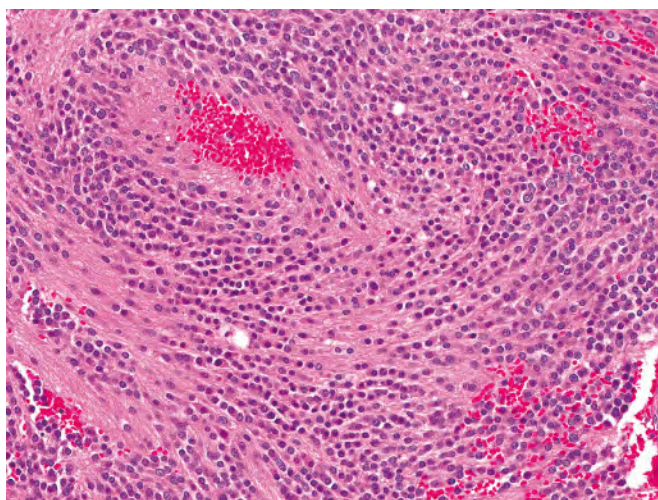


Fig. 16.13 Olfactory neuroblastoma is composed of small monotonous cells with round nuclei in an eosinophilic fibrillary stroma. The cells appear to swirl around a blood-filled vascular channel.

Olfactory Neuroblastoma

Olfactory neuroblastomas arise from neuroectodermal olfactory cells within the mucosa, particularly in the superior aspect of the nasal cavity and cribriform plate. It can occur at any age with a peak in the fifth and sixth decades. Patients typically present with nasal obstruction and/or epistaxis. Histologically, olfactory neuroblastomas are *small round blue cell neoplasms*, a category of morphologically similar tumors including lymphoma, small cell carcinoma, Ewing sarcoma and rhabdomyosarcoma. Olfactory neuroblastomas are typically composed of well-circumscribed nests and lobules of cells separated by fibrovascular stroma (Fig. 16.13). A fibrillary matrix, representing tangled neuronal cell processes, is often present. Consistent with their

neuroendocrine origin, electron microscopy demonstrates membrane-bound secretory granules, and immunohistochemistry is positive for neuroendocrine markers such as synaptophysin, chromogranin, INSM1, and CD56. Depending on tumor stage and grade, combinations of surgery, radiation therapy, and chemotherapy yield 5-year survival rates of 40% to 90%. Genetic changes in olfactory neuroblastoma are heterogeneous, with cases demonstrating varying chromosomal gains and/or losses.

NUT Carcinoma

This rare tumor occurs most commonly in the head and neck (sinonasal tract, nasopharynx, salivary gland) but can also arise in other anatomic sites including the thorax or abdomen. It can occur at any age, from infancy to late adulthood, but the true incidence is not known, as it histologically resembles SCC and is likely frequently misdiagnosed. NUT carcinoma is clinically distinctive due to its extremely aggressive behavior and resistance to conventional therapy; most patients survive for less than 1 year following diagnosis. It is uniformly associated with translocations that create fusion genes encoding chimeric proteins composed of most of NUT, a chromatin regulator named “nuclear protein in testis,” and a portion of a “chromatin reader” protein, usually BRD4. Drugs that displace BRD4-NUT from chromatin induce NUT carcinoma cells to terminally differentiate, and targeted therapy with inhibitors of chromatin readers is being tested clinically.

Nasopharyngeal Carcinoma

Nasopharyngeal carcinoma is characterized by a distinctive geographic distribution, a close anatomic relationship to lymphoid tissue, and an association with EBV infection. The nomenclature for nasopharyngeal carcinoma continues to evolve, but, at present, is described as one of three histologic subtypes: (1) keratinizing SCC, (2) nonkeratinizing SCC, and (3) basaloid SCC. *Lymphoepithelioma* or *lymphoepithelial carcinoma* are historic names for this tumor that are no longer used.

The etiology and epidemiology of nasopharyngeal carcinomas is complex. Peak geographic locations include parts of southern China, northern Africa, southeast Asia, and Inuit regions of Alaska and Canada. EBV infection, diets high in nitrosamines (which are found in fermented foods and salted fish), and environmental insults (including smoking and chemical fumes) have been linked to the development of these tumors. In the nonkeratinizing form, most patients have antibodies against EBV early antigens or viral capsid antigens, and PCR can be used to detect EBV DNA in the serum.

MORPHOLOGY

Histologically, keratinizing and nonkeratinizing nasopharyngeal carcinomas resemble well-differentiated and poorly differentiated SCCs arising in other locations. The nonkeratinizing subtype is composed of large epithelial cells with oval or round vesicular nuclei, prominent nucleoli, and indistinct cell borders; the latter impart the appearance of a syncytium (Fig. 16.14B). Abundant, normal-appearing lymphocytes, predominantly T cells, infiltrate the tumors. EBV-encoded RNAs such as EBER-1 or proteins such as LMP-1 can be identified in the malignant epithelial cells by *in situ* hybridization (Fig. 16.14C) or immunohistochemistry, respectively.

Primary nasopharyngeal carcinomas often present at advanced stages with nasal obstruction, epistaxis, and metastases to the ipsilateral cervical lymph nodes in up to 70% of patients. Radiotherapy is the standard treatment and results in an overall 5-year survival of approximately 60%. Outcome, however, varies with histology; the stage-dependent 5-year survival for nonkeratinizing nasopharyngeal

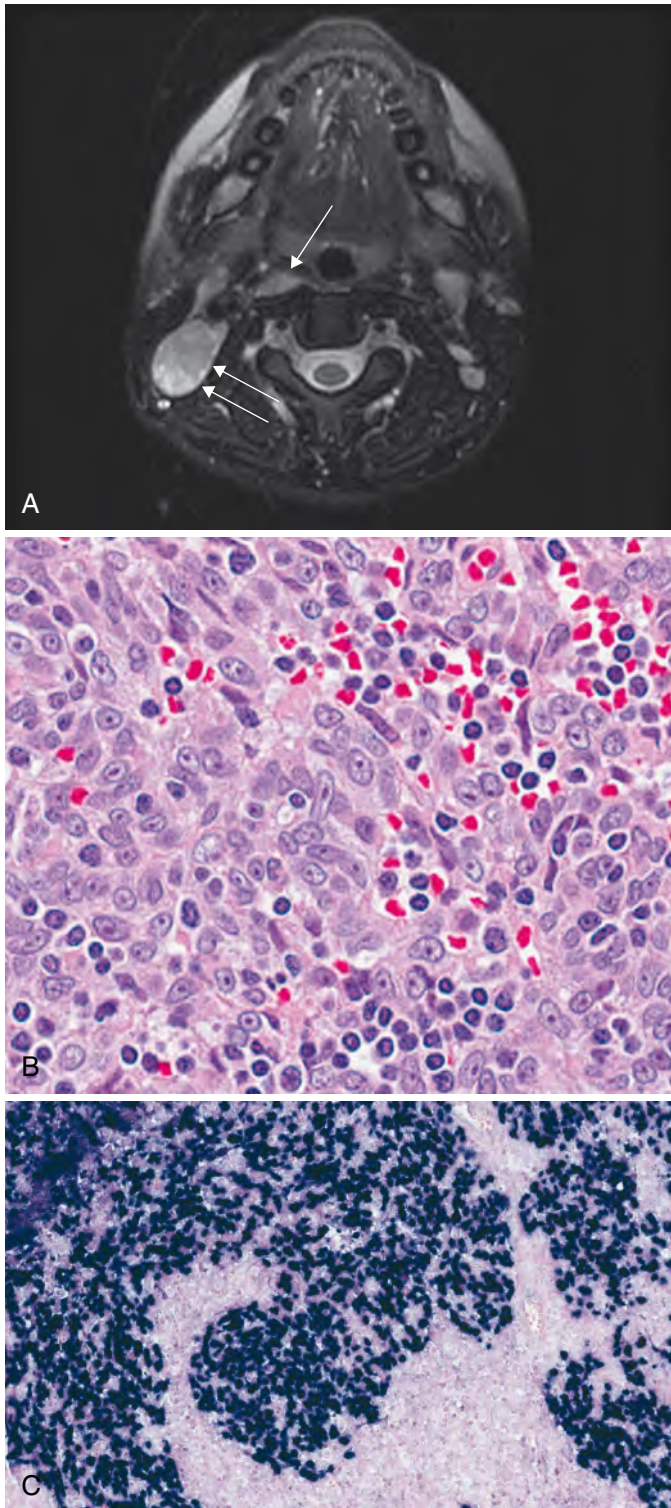


Fig. 16.14 Nasopharyngeal carcinoma, nonkeratinizing subtype. (A) Computed tomography study demonstrating thickening of the nasopharyngeal region (*arrow*) and an enlarged cervical lymph node (*double arrow*). (B) The syncytial clusters of malignant epithelium are infiltrated by benign lymphocytes. (C) In situ hybridization for EBER-1, a small nuclear RNA encoded by Epstein-Barr virus.

carcinoma ranges from 70% to 98%, whereas for the keratinizing form 5-year survival is only 20%. These stark differences are attributable to divergent therapeutic responses, as undifferentiated carcinomas are substantially more radiosensitive than keratinizing carcinomas. Checkpoint inhibitors can be used to treat advanced nasopharyngeal carcinomas, including recurrent cancer that cannot be treated with surgery or radiation, or that has spread to other parts of the body.

LARYNX

The most common disorders of the larynx are inflammatory. Tumors are uncommon but are amenable to resection, though often at the price of loss of natural voice.

Inflammatory Lesions

Laryngitis may occur as the sole manifestation of allergic, viral, bacterial, or chemical insult, but it is more commonly associated with generalized upper respiratory tract infection, heavy environmental toxin exposure (e.g., tobacco smoke), or gastroesophageal reflux due to the irritation caused by gastric contents. The larynx may also be affected in systemic infections, such as tuberculosis and diphtheria. Although most infections are self-limited, the sequelae can be serious, especially in infancy or childhood, when mucosal congestion, exudation, or edema may cause laryngeal obstruction. In particular, laryngoepiglottitis caused by respiratory syncytial virus, *Haemophilus influenzae*, or β -hemolytic streptococci may induce such sudden swelling of the epiglottis and vocal cords in the small airways of infants and young children as to constitute a medical emergency. This is uncommon in adults because of the larger size of the larynx as well as stronger accessory respiratory muscles. Fortunately, effective vaccines against *H. influenzae* and respiratory syncytial virus substantially decrease the chances of this serious complication in children. *Croup*, once thought to describe diphtheria, now generally refers to a viral laryngotracheobronchitis in children that produces a characteristic inspiratory stridor due to airway narrowing. Finally, laryngitis that is common in heavy smokers predisposes to squamous epithelial metaplasia and laryngeal carcinoma.

Reactive Nodules (Vocal Cord Nodules and Polyps)

Reactive nodules, also called polyps, develop on the vocal cords, most often in heavy smokers or those who impose great strain on their vocal cords (so-called *singer's nodules*) (Fig. 16.15). Because of

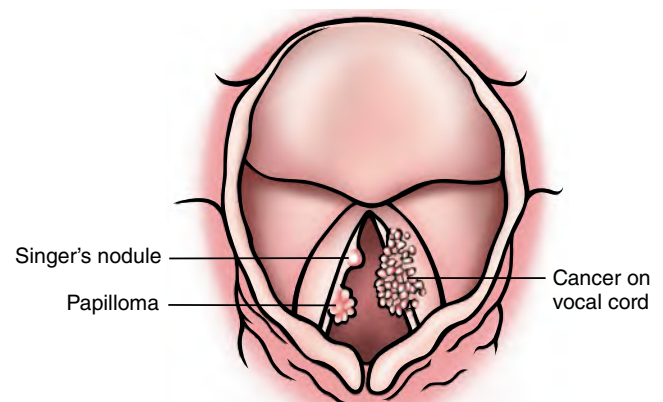


Fig. 16.15 Diagrammatic comparison of a singer's nodule, a benign papilloma, and an exophytic carcinoma of the larynx to highlight differences in their clinical appearance.

their location and accompanying inflammation, they characteristically change the character of the voice and often cause progressive hoarseness. Despite this, the risk of developing cancer in these lesions is almost nonexistent. By convention, vocal cord nodules are bilateral and polyps are unilateral. Adults are most often affected. Both nodules and polyps are smooth, rounded, sessile, or pedunculated excrescences, generally only a few millimeters in greatest dimension, and usually located on the true vocal cords. The histologic features of nodules and polyps are similar; they are covered by squamous epithelium that may become hyperkeratotic, hyperplastic, or even slightly dysplastic overlying a loose myxoid connective tissue stroma. The latter may be variably fibrotic, fibrinous, or highly vascularized. Nodules on opposing vocal cords may impinge on each other and cause ulceration.

Squamous Papilloma and Papillomatosis

Laryngeal squamous papillomas are benign neoplasms, usually located on the true vocal cords, that form soft, exophytic proliferations that rarely exceed 1 cm in diameter (see Fig. 16.15). The growths are composed of multiple slender, finger-like projections supported by central fibrovascular cores and covered by an orderly stratified squamous epithelium. When papillomas develop on the free edge of a vocal cord, trauma may lead to ulceration that can be accompanied by hemoptysis.

Papillomas are most often solitary in adults but can be multiple in both children and adults, a condition referred to as *recurrent papillomatosis*. The lesions are caused by HPV types 6 and 11, acquired via person-to-person contact with an HPV-positive individual or via the maternal birth canal. They frequently recur but rarely transform to SCC, mainly in the context of radiation exposure or smoking. Treatment of papillomas usually consists of endoscopic resection. HPV vaccination against types 6 and 11 has reportedly decreased the incidence of papillomatosis.

Carcinoma of the Larynx

Carcinoma of the larynx is predominantly a squamous cell carcinoma seen in adult chronic smokers.

Hyperplasia-Dysplasia-Carcinoma Sequence

Epithelial alterations in the larynx range from hyperplasia to dysplasia, carcinoma in situ, and invasive carcinoma. Grossly, epithelial surfaces may be smooth, white or reddened with focal thickenings, roughened by keratosis, or punctuated by irregular verrucous or ulcerated white-pink lesions (see Figs. 16.15 and 16.16).

Nondysplastic hyperplasias have almost no potential for malignant transformation. For lesions with dysplasia, the risk of malignant transformation is directly proportional to the initial grade of dysplasia. For example, the risk of transformation is only 1% to 2% over 5 to 10 years for lesions with mild dysplasia but increases to 5% to 10% for those with severe dysplasia. Histologic evaluation is the only way to grade dysplasia.

Laryngeal carcinoma is most often related to tobacco smoke; risk is proportional to exposure. Prior to malignant transformation, epithelial changes frequently regress after smoking cessation. Alcohol synergizes with tobacco to increase risk substantially. Nutritional factors, asbestos exposure, irradiation, and high-risk HPV infection can also increase cancer risk.

those at other sites, beginning as in situ lesions that become pearly gray, wrinkled mucosal plaques and, ultimately, ulcerated fungating masses (see Fig. 16.16). Histologically, the degree of anaplasia is highly variable and can include tumor giant cells and bizarre mitotic figures. As expected of sites in which neoplasia arises following recurrent exposure to environmental carcinogens, squamous hyperplasia with foci of dysplasia or carcinoma in situ may be present within adjacent mucosae.

Laryngeal carcinoma presents most commonly in the sixth decade of life. The initial manifestation is often persistent hoarseness, dysphagia, and dysphonia. Prognosis is directly related to clinical stage. In early disease, organ-preserving techniques (e.g., laser surgery, microsurgery, and radiation therapy) are being used with greater frequency. Combined chemotherapy and radiation therapy, with or without laryngectomy, is typically required for more advanced or recurrent disease.

KEY CONCEPTS

Upper Airways

- Rhinitis can be infectious or allergic in nature and, over time, can lead to chronic rhinitis, sinusitis, and development of nasal polyps.
- Pharyngitis and tonsillitis are typically caused by upper respiratory tract infection by rhinoviruses, echoviruses, and adenoviruses.
- Nasopharyngeal carcinoma is most often caused by EBV; tumors are most common among African children and Asian adults.
- Laryngitis can be caused by a host of etiologies, including allergic, viral, bacterial, and chemical insults.
- Vocal cord nodules and polyps are reactive lesions seen in smokers or individuals who strain their vocal cords.
- Laryngeal SCC is linked to smoking and alcohol use and is most prevalent in older males.



Fig. 16.16 Laryngeal carcinoma. Note the large, ulcerated, fungating lesion involving the right vocal cord and pyriform sinus.

MORPHOLOGY

About 95% of laryngeal carcinomas are SCCs. Most arise on the vocal cords, but they may develop above or below the cords and involve the epiglottis, aryepiglottic folds, or pyriform sinuses. SCCs of the larynx develop similarly to

EARS

Although they rarely shorten life, disorders of the ear often impact its quality. The most common aural disorders, in descending order of frequency, are (1) acute and chronic otitis (most often involving the middle ear and mastoid), sometimes leading to a cholesteatoma; (2) symptomatic otosclerosis; (3) aural polyps; (4) labyrinthitis; (5) carcinomas, largely of the external ear; and (6) paragangliomas, mostly in the middle ear. Only those conditions that have distinctive morphologic features (save for labyrinthitis) are described. Paragangliomas are discussed later.

INFLAMMATORY LESIONS

Acute and chronic otitis media occur most often in infants and children. The origin is typically viral infection that induces a serous exudate. Superimposed bacterial infection, most frequently by *Streptococcus pneumoniae*, nontypeable *H. influenzae*, or *Moraxella catarrhalis*, can lead to suppurative inflammation.

Repeated bouts of acute otitis media with failure of resolution lead to chronic disease. The causative agents are usually *Pseudomonas aeruginosa*, *S. aureus*, fungus, or, in some cases, a polymicrobial infection. Chronic infection has the potential to perforate the eardrum, encroach on the ossicles or labyrinth, spread into the mastoid spaces, and even penetrate the cranial vault to produce temporal cerebritis or abscess. In individuals with diabetes, otitis media caused by *P. aeruginosa* is especially aggressive and can spread widely, resulting in destructive necrotizing otitis media.

Cholesteatomas are nonneoplastic, cystic lesions associated with chronic otitis media. The cysts are typically 1 to 4 cm in diameter, lined by keratinizing squamous or metaplastic mucus-secreting epithelium, and filled with amorphous, keratin debris. Cholesterol spicules may be present. Although the pathogenesis is not clear, it is thought that chronic inflammation and perforation of the eardrum, with ingrowth of the squamous epithelium or metaplasia of the secretory epithelial lining of the middle ear, promote formation of a squamous cell nest that becomes cystic. Cyst rupture induces a brisk inflammatory reaction that includes giant cells engulfing partially necrotic squamous cells and other particulate debris. These lesions, by progressive enlargement, can erode into the ossicles, labyrinth, adjacent bone, or surrounding soft tissue and sometimes produce visible neck masses.

OTOSCLEROSIS

Otosclerosis refers to abnormal bone deposition in the middle ear about the rim of the oval window into which the footplate of the stapes fits. Both ears are usually affected. At first, there is fibrous ankylosis of the footplate. This often occurs in the early decades of life. Over time, bony overgrowth anchors the footplate into the oval window. The severity of the hearing loss reflects the degree of immobilization. Minimal degrees of otosclerosis are common in young to middle-aged adults, but fortunately more severe symptomatic disease is relatively rare. Otosclerosis is familial in most instances, with an autosomal dominant transmission and variable penetrance. The basis for the osseous overgrowth is obscure, but it appears to represent uncoupling of normal bone resorption and bone formation. Thus, it begins with bone resorption, followed by fibrosis and vascularization of the temporal bone in the immediate vicinity of the oval window. Over time the fibrosis is replaced by dense new bone anchoring the footplate of the stapes. In most instances, the process is slowly progressive over the span of decades and can eventually lead to marked hearing loss.

TUMORS

Epithelial and mesenchymal tumors of the external, middle, or internal ear are rare, except for basal cell carcinomas or SCCs arising on the pinna (external ear). These tend to occur in elderly males and are associated with sun exposure. In contrast, SCCs of the ear canal occur most often in middle-aged to elderly females and are not associated with sun exposure. SCCs involving the ear resemble their counterparts at other skin locations, beginning as papules that extend and eventually erode and invade. Despite local invasion, basal cell carcinomas and SCCs involving the pinna rarely spread. The outlook is more bleak when SCCs arise in the external ear canal, as these may invade the cranial cavity or metastasize to regional lymph nodes, accounting for a 5-year survival of only about 50%.

KEY CONCEPTS

Ears

- Infections of the ear are common in children and typically viral in etiology. Chronic infections can be complicated by bacterial infections, which, in turn, can lead to secondary complications including perforated eardrum, as well as spread to the ossicles or mastoid spaces.
- Otosclerosis, with its associated hearing loss, is caused by the abnormal deposition of bone in the middle ear.

NECK

Most of the conditions that involve the neck are described elsewhere (e.g., primary squamous cell and basal cell carcinomas of the skin, melanomas, lymphomas) or are components of systemic disorders (e.g., generalized rashes, lymphadenopathy of infectious mononucleosis, tonsillitis). What remains to be considered here are a few uncommon lesions unique to the neck.

BRANCHIAL CLEFT CYST (CERVICAL LYMPHOEPITHELIAL CYST)

Most branchial cleft cysts are thought to arise from remnants of the second branchial arch and are most commonly observed in young adults between 20 and 40 years of age. These benign lesions usually appear on the upper lateral aspect of the neck along the sternocleidomastoid muscle. The slowly enlarging cysts are well-circumscribed, 2 to 5 cm in diameter, and usually lined by stratified squamous or pseudostratified columnar epithelium. The fibrous cyst walls typically contain lymphoid tissue with prominent germinal centers. Cyst contents may be clear and watery or mucinous and may also contain desquamated cells and granular cellular debris. The cysts are readily excised. Similar lesions may appear in the parotid gland or oral cavity beneath the tongue. Branchial cleft cysts usually do not undergo malignant transformation. Most cystic SCCs in the neck are metastases from cancers of the upper aerodigestive tract.

THYROGLOSSAL DUCT CYST

The thyroid anlage begins in the region of the foramen cecum at the base of the tongue; as the gland develops, it descends to its definitive midline location in the anterior neck. Remnants of this developmental tract may persist, resulting in 1- to 4-cm cysts that are lined by stratified squamous epithelium when located near the base of the tongue or by pseudostratified columnar epithelium in lower locations. Transitional epithelial differentiation patterns also occur. The fibrous cyst

wall often includes lymphoid aggregates or thyroid remnants. Malignant transformation of the lining epithelium is exceedingly rare. The definitive treatment is surgical excision (Sistrunk procedure).

PARAGANGLIOMA

Parangliomas arise from neuroendocrine cells associated with the sympathetic and parasympathetic nervous systems and occur at many sites. Adrenal medullary pheochromocytomas are the most common paraganglioma, accounting for 80% to 85% of all cases, adrenal and extra-adrenal (Chapter 24). Approximately 70% of extra-adrenal paragangliomas occur in the head and neck. Loss-of-function mutations in genes encoding succinate dehydrogenase subunits (which are also common in gastrointestinal stromal tumors) or cofactors in mitochondrial oxidative phosphorylation occur frequently in both hereditary and spontaneous paragangliomas. How these mutations contribute to tumor development is not yet clear, but it is suspected that they do so by altering cellular metabolism, which you will recall is one of the hallmarks of neoplasia (Chapter 7). Interestingly, the incidence of these tumors is greater in people living at high altitudes.

Parangliomas typically develop in two locations:

- *Paravertebral paraganglia* (e.g., organ of Zuckerkandl). Such tumors have sympathetic connections and stain positively for chromaffin, which detects cells producing catecholamines.
- *Near the great vessels of the head and neck*, the so-called *aortico-pulmonary chain*, including the *carotid bodies* (i.e., carotid body tumor); aortic bodies; jugulotympanic ganglia (sometimes referred to as *glomus tympanicum* or *jugulare*); ganglion nodosum of the vagus nerve; and clusters located about the oral cavity, nose, nasopharynx, larynx, and orbit. These are innervated by the parasympathetic nervous system and only rarely produce catecholamines.

MORPHOLOGY

The **carotid body tumor** is a typical parasympathetic paraganglioma. It rarely exceeds 6 cm in diameter and arises close to or envelops the bifurcation of the common carotid artery. The tumor tissue is red-pink to brown, and the microscopic features are characteristic of paragangliomas at all sites. Nests (zellballen) of round to oval **chief cells** with abundant clear or granular eosinophilic cytoplasm and uniform, round to ovoid, sometimes vesicular, nuclei are surrounded by delicate vascular septae (Fig. 16.17). There is little cellular pleomorphism, and mitoses are scant. The chief cells, which are neuroectodermal in origin, stain strongly for neuroendocrine markers including chromogranin, synaptophysin, INSM1 (insulinoma-associated protein 1), and CD56. The supporting network of spindle-shaped stromal cells, collectively called **sustentacular cells**, are positive for S-100 protein. Electron microscopy often detects well-demarcated neuroendocrine granules in paravertebral tumors, but the number of these granules is variable, and they tend to be scant in nonfunctioning tumors.

Parangliomas (PGLs) are rare, slow-growing, painless masses that usually arise in the fifth and sixth decades of life. They commonly occur singly and sporadically but also may be familial. Tumors associated with multiple endocrine neoplasia type 2 most often occur within the adrenal (pheochromocytomas) (Chapter 24), whereas tumors associated with hereditary paraganglioma syndromes (PGL types 1 to 4, involving the succinate dehydrogenase genes) typically involve the head and neck. Approximately one-third of head and neck paragangliomas are thought to be associated with germline mutations. *Carotid body tumors* frequently recur after incomplete resection and, despite their benign appearance, may metastasize to regional lymph nodes and

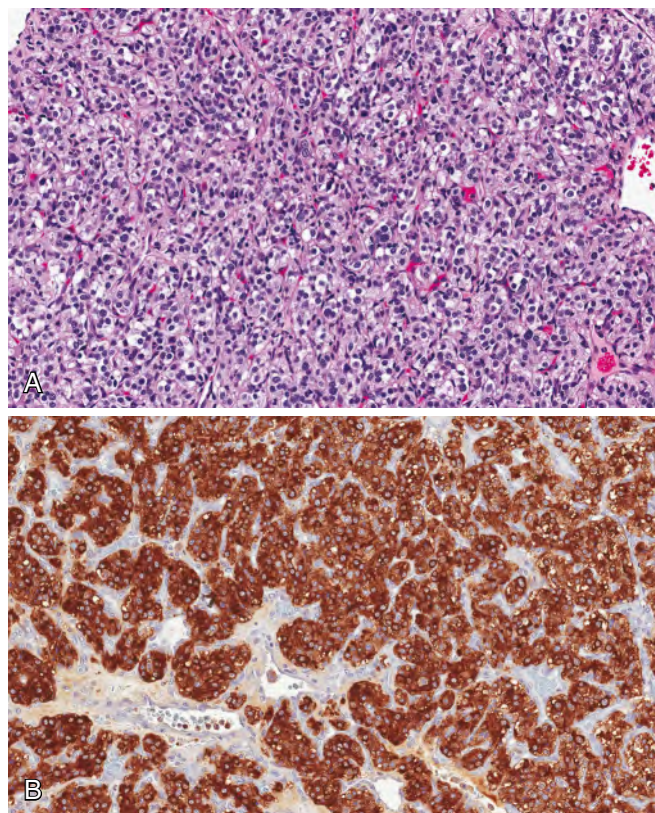


Fig. 16.17 Carotid body tumor. (A) Low-power view showing tumor clusters separated by septa (zellballen). The septae are marked by capillaries containing red cells. (B) Immunohistochemistry demonstrating positivity for chromogranin in the tumor cells.

distant sites. Up to 50% of metastatic paragangliomas ultimately prove fatal, largely because of infiltrative growth. Unfortunately, histologic features do not predict the clinical course—mitoses, pleomorphism, and even vascular invasion are not reliable indicators. However, succinate dehydrogenase B gene (*SDHB*) mutations are associated with the highest rates of metastasis (30% to 50%).

KEY CONCEPTS

Neck

- Branchial cleft cysts arise from the region of the second branchial pouch and typically present in young adults.
- Thyroglossal duct cysts arise due to incomplete descent of the thyroid analog from the foramen cecum in the base of the tongue.
- Seventy percent of extra-adrenal paragangliomas occur in the head and neck. These tumors are most common in the fifth and sixth decades of life and can be hereditary or sporadic. Hereditary head and neck paragangliomas are associated with succinate dehydrogenase gene mutations, occur at younger ages, are often multiple, and can metastasize.

SALIVARY GLANDS

There are three major salivary glands, parotid, submandibular, and sublingual, as well as innumerable minor salivary glands distributed throughout the upper aerodigestive tract; all may be involved by inflammatory or neoplastic disease.

XEROSTOMIA

Xerostomia is defined as a dry mouth resulting from a decrease in the production of saliva. Its incidence has been reported to be as high as 20% in individuals older than 70 years of age in some populations. Xerostomia is characteristic of the autoimmune disorder *Sjögren syndrome* (Chapter 6), which usually also leads to dry eyes (keratoconjunctivitis sicca). Lack of salivary secretions is also a major complication of radiation therapy for head and neck cancers. Xerostomia is, however, most often a side-effect of commonly prescribed medications, including anticholinergic, antidepressant/antipsychotic, diuretic, antihypertensive, sedative, muscle relaxant, analgesic, and antihistamine drugs. Xerostomia may present as dry mucosa and/or atrophy of the papillae of the tongue, with fissuring and ulcerations. In Sjögren syndrome, inflammatory enlargement of the salivary glands may also occur. Complications of xerostomia include increased rates of dental caries, candidiasis, and difficulty in swallowing and speaking.

INFLAMMATION (SIALADENITIS)

Sialadenitis may be induced by trauma, viral or bacterial infection, or autoimmune disease. A common manifestation of sialadenitis is the **mucocoele**. The most common viral cause of sialadenitis is **mumps**, which affects the major salivary glands, particularly the parotids, and frequently involves other glandular organs such as the pancreas and testes (Chapter 8).

Mucocoele

Mucocoeles are most often found on the lower lip as the result of trauma (Fig. 16.18A), occur at all ages, and present as fluctuant lower lip swellings with a blue translucent hue. This common salivary gland lesion results from either blockage or rupture of a salivary gland duct, with consequent leakage of saliva into the surrounding connective tissue stroma. Patients may report a history of changes in the size of the lesion, particularly in association with meals. Histologically, mucocoeles are pseudocysts lined by granulation tissue or fibrous connective tissue and filled with mucin and inflammatory cells,

particularly macrophages (Fig. 16.18B). Complete excision of the cyst and accompanying minor salivary gland lobule is required, as incomplete excision may lead to recurrence.

Ranula is a term reserved for epithelial-lined cysts that arise when the duct of the sublingual gland has been damaged. A ranula may become so large that it develops into a *plunging ranula*, a colorful description of a cyst that has dissected through the connective tissue stroma connecting the two bellies of the mylohyoid muscle.

Sialolithiasis and Nonspecific Sialadenitis

Nonspecific bacterial sialadenitis, involving the submandibular and other major salivary glands, is common and most often caused by infection with *S. aureus* and *Streptococcus viridans* following ductal obstruction by stones (sialolithiasis). Stone formation is sometimes related to obstruction of salivary gland orifices by impacted food debris or local edema after injury, but no underlying cause is detected in many cases. Reduced secretion of saliva, as sometimes occurs in patients receiving long-term phenothiazines, may predispose to bacterial infection. Similarly, decreased salivary secretions due to dehydration may facilitate bacterial suppurative parotitis, particularly in elderly patients with a recent history of major thoracic or abdominal surgery.

Obstructive processes and bacterial invasion lead to inflammation of the affected glands that is often interstitial or, when induced by staphylococcal or other pyogenes, associated with suppurative necrosis and abscess formation. Inflammation causes painful enlargement and, sometimes, a purulent ductal discharge. Disease is almost always unilateral and involves a single gland.

NEOPLASMS

Despite their relatively simple morphology, the salivary glands give rise to more than 30 histologically distinct tumors. A classification and the relative incidence of benign and malignant tumors is listed in Table 16.5; not included are the rare benign and malignant mesenchymal neoplasms. Despite this diversity, a small number of neoplasms makes up more than 90% of salivary gland tumors. These will be the focus of the following discussion.

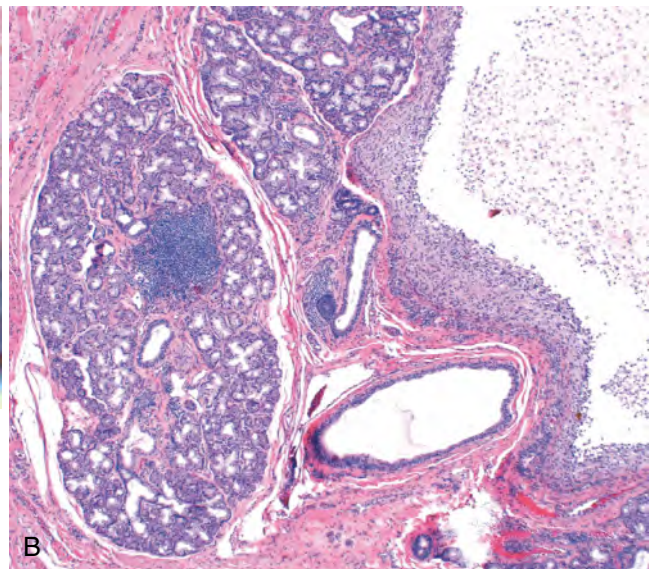


Fig. 16.18 Mucocoele. (A) Fluctuant fluid-filled lesion on the lower lip subsequent to trauma. (B) Cystlike cavity filled with mucinous material and lined by organizing granulation tissue adjacent to intact salivary gland.

TABLE 16.5 Histologic Classification and Incidence of the Most Common Benign and Malignant Tumors of the Salivary Glands

Benign	Malignant
Pleomorphic adenoma (mixed tumor)	Mucoepidermoid carcinoma
Warthin tumor	Acinic cell carcinoma
Oncocytoma	Adenoid cystic carcinoma
Canalicular adenoma	Salivary duct carcinoma
Basal cell adenoma	Carcinoma ex pleomorphic adenoma
Other adenomas	Other carcinomas

Salivary gland neoplasms represent less than 2% of all tumors in humans. About 65% to 80% arise within the parotid gland, 10% in the submandibular gland, and the remainder in the minor salivary glands, including the sublingual glands. Most parotid tumors are benign, but 40% of submandibular, 50% of minor salivary gland, and 70% to 90% of sublingual tumors are malignant. Thus, in general terms, the malignant potential of salivary gland tumors is inversely proportional to gland size, with smaller glands harboring greater numbers of cancers.

Salivary gland tumors occur predominantly in adults, with only approximately 5% occurring before 16 years of age. Parotid gland neoplasms produce distinctive swellings in front of and below the ear. Benign tumors are often present for many months to several years before coming to clinical attention, whereas high-grade cancers are often detected more quickly due to their rapid growth. Most tumors are mobile on palpation except for highly invasive carcinomas. Despite these generalities, there are no reliable clinical criteria to differentiate benign from malignant lesions, and biopsy and/or surgical resection is required for definitive diagnosis.

Pleomorphic Adenoma

Pleomorphic adenomas are the most common salivary gland neoplasms. They represent about 60% of tumors in the parotid, are less common in the submandibular glands, and are relatively rare in the minor salivary glands. Pleomorphic adenomas are benign tumors that consist of a mixture of ductal (epithelial), myoepithelial, and mesenchymal cells, which explain why they are also termed *mixed tumors*.

Little is known about the origin of pleomorphic adenomas, but radiation exposure increases risk. It is hypothesized that all tumor elements, including those that appear mesenchymal, are of myoepithelial or ductal reserve cell (stem cell) origin. Many cases are associated with chromosomal rearrangements that induce overexpression of PLAG1, a transcription factor that promotes expression of genes that increase cell growth, including components of growth factor receptor signaling pathways. Mutations of the *HMGA2* gene, which encodes a DNA-binding protein, are associated with many of the cases that lack PLAG overexpression.

MORPHOLOGY

Most pleomorphic adenomas present as rounded, well-demarcated masses that rarely exceed 6 cm in greatest dimension (Fig. 16.19). Although circumscribed (Fig. 16.20A), in some locations (particularly the palate) the capsule of pleomorphic adenomas is not fully developed, and expansile growth produces **tonguelike protrusions** into the surrounding gland. These may lead to recurrences if the tumor is merely enucleated. The cut surface is gray-white with myxoid and blue translucent areas of chondroid (cartilage-like) stroma.

The dominant histologic feature is morphologic heterogeneity (not true nuclear pleomorphism). The epithelial elements resembling ductal cells or myoepithelial cells are arranged as ducts, acini, irregular tubules, strands, or sheets of cells. These elements are typically dispersed within a background of **chondromyxoid matrix**, consisting of loose myxoid and hyaline tissue containing

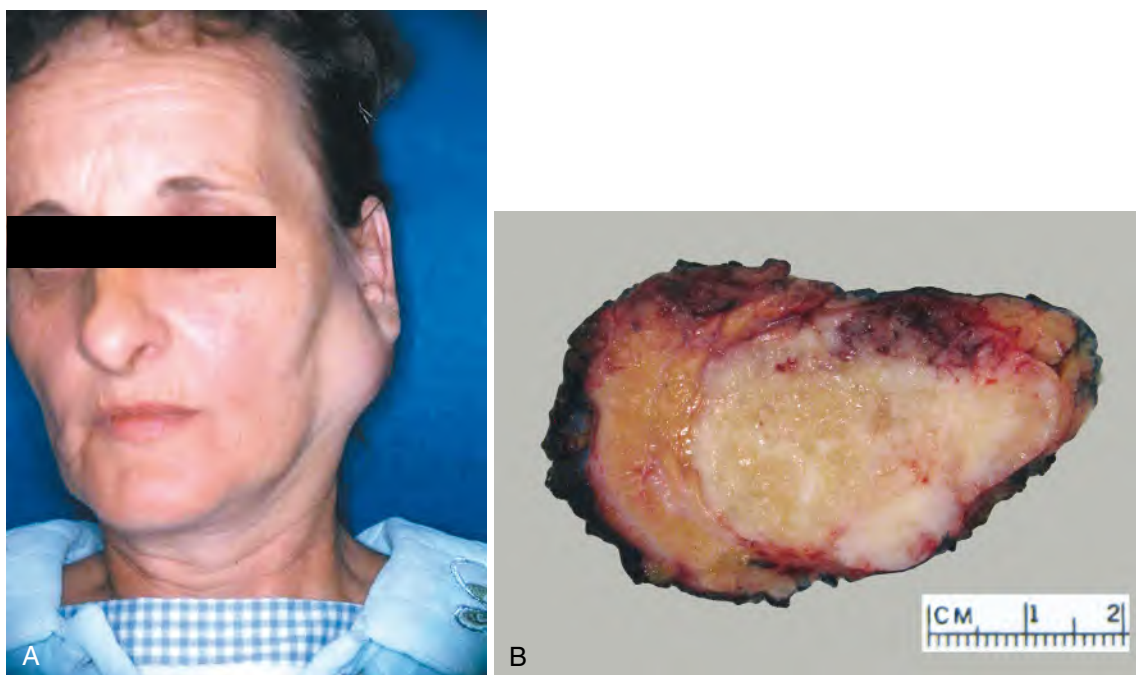


Fig. 16.19 Pleomorphic adenoma. (A) Slowly enlarging neoplasm in the parotid gland of many years' duration. (B) Grossly, this representative cross-section of a parotidectomy specimen shows a circumscribed, yellow-white tumor and adjacent normal salivary gland tissue on the left.

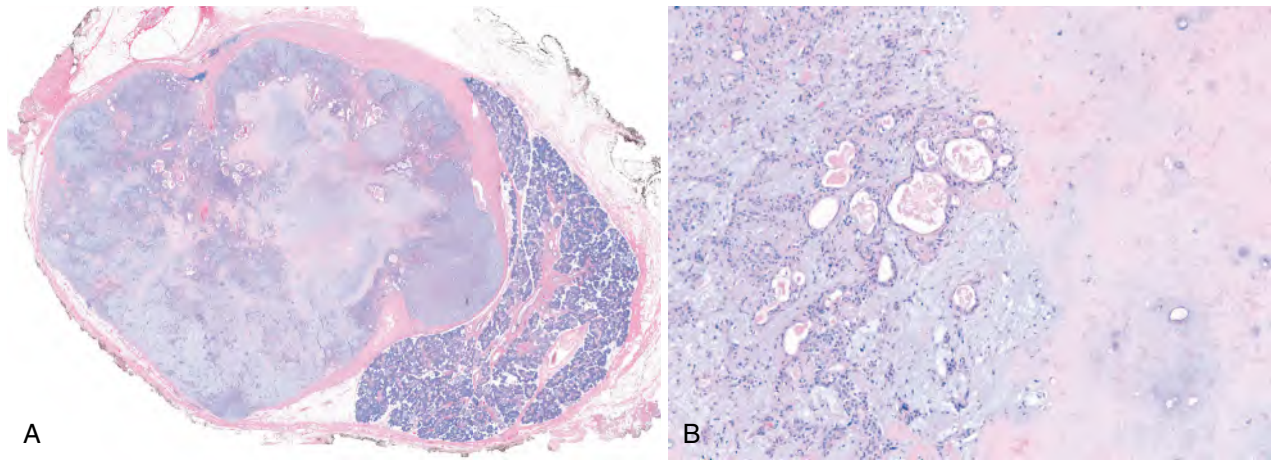


Fig. 16.20 Pleomorphic adenoma. (A) Low-power histologic view showing a well-demarcated tumor with adjacent normal salivary gland parenchyma on the right. (B) High-power view showing epithelial and myoepithelial cells forming ducts (*left*) and a chondromyxoid matrix (*right*).

islands of cartilage and, rarely, foci of bone (Fig. 16.20B). Sometimes the epithelial cells form well-developed ducts lined by cuboidal to columnar cells with an underlying layer of deeply chromatic, small myoepithelial cells. In other instances, there may be strands or sheets of myoepithelial cells. Islands of well-differentiated squamous epithelium may also be present. In most cases, there is no epithelial dysplasia or mitotic activity. Tumors behave similarly whether they are primarily composed of epithelial or mesenchymal elements.

Pleomorphic adenomas present as painless, slow-growing, mobile, discrete masses within the parotid or submandibular areas or in the buccal cavity. About 4% recur following parotidectomy months to years later. In contrast, recurrence after simple enucleation approaches 25%, likely due to tumor extension beyond the capsule.

Rates of malignant transformation in pleomorphic adenoma (carcinoma ex pleomorphic adenoma) correlate with age of the lesion; 2% of tumors that have been present for less than 5 years harbor cancers, but this increases to 10% for those present for more than 15 years. Cancers are usually adenocarcinomas or undifferentiated carcinomas. They are highly infiltrative and may completely replace the precursor lesion, making it difficult to substantiate the diagnosis of carcinoma arising in pleomorphic adenoma. Regrettably, when they appear, these cancers are among the most aggressive of all salivary gland neoplasms, with mortality rates of 30% to 50% at 5 years.

Warthin Tumor (Papillary Cystadenoma Lymphomatosum)

The Warthin tumor is the second most common salivary gland neoplasm. It is unique in that it arises almost exclusively in the parotid gland. Warthin tumors are benign and occur more commonly in males than in females, usually in the fifth to seventh decades of life. The risk is increased eight-fold in smokers. Most Warthin tumors are unifocal, but about 10% are multifocal and 10% bilateral.

MORPHOLOGY

Warthin tumors are round to oval encapsulated masses, 2 to 5 cm in diameter, and readily palpable within the superficial parotid gland. Transection reveals a pale gray surface punctuated by narrow cystic or cleftlike spaces filled with mucinous or serous secretions and frequently narrowed by polypoid projections of lymphoepithelial elements (Fig. 16.21A). The lining is composed of a double layer of onco-

cytic cells; the innermost layer is columnar, while cuboidal cells occupy the outer layer (Fig. 16.21B). The term **oncocyte** refers to large cells containing numerous mitochondria that impart a granular, eosinophilic appearance to the cytoplasm. Secretory cells dispersed in the inner layer account for secretions within the dilated lumen. Foci of squamous metaplasia may be present.

The histogenesis of Warthin tumors is debated; it is not clear whether the epithelium is truly neoplastic (monoclonal) or is a reactive proliferation (polyclonal). Regardless, the epithelial cells are thought to secrete chemoattractants for the lymphoid cells, which are reactive. Although Warthin tumors have occasionally arisen within cervical lymph nodes, these neoplasms are benign and have low recurrence rates of only 2% after resection.

Mucoepidermoid Carcinoma

Mucoepidermoid carcinoma is the most common primary malignant tumor of salivary glands, representing about 15% of all salivary gland tumors. Most (60% to 70%) occur in the parotids, but mucoepidermoid carcinomas account for a large fraction of salivary gland neoplasms in the other glands, particularly the minor salivary glands. More than one-half of cases are associated with a balanced (11;19) (q21;p13) chromosomal translocation that creates a fusion gene composed of portions of the *CRTC1* and *MAML2* genes. The part of *MAML2* that is fused cannot interact with Notch signaling components. The *CRTC1::MAML2* fusion protein is believed to play a key role in the genesis of this tumor, possibly by perturbing Notch and cAMP-dependent signaling pathways.

MORPHOLOGY

Mucoepidermoid carcinomas can grow as large as 8 cm in diameter. Although they appear circumscribed, well-defined capsules are not present and the tumors are often infiltrative. On transection, mucoepidermoid carcinomas are pale, gray-white and frequently contain small, mucin-containing cysts. Histology demonstrates cords, sheets, or cystic configurations of squamous, mucous, or intermediate cells (Fig. 16.22A,B). The cytology may be monotonous and bland or, alternatively, highly anaplastic and unmistakably malignant. Accordingly, mucoepidermoid carcinomas are subclassified as low, intermediate, or high grade.

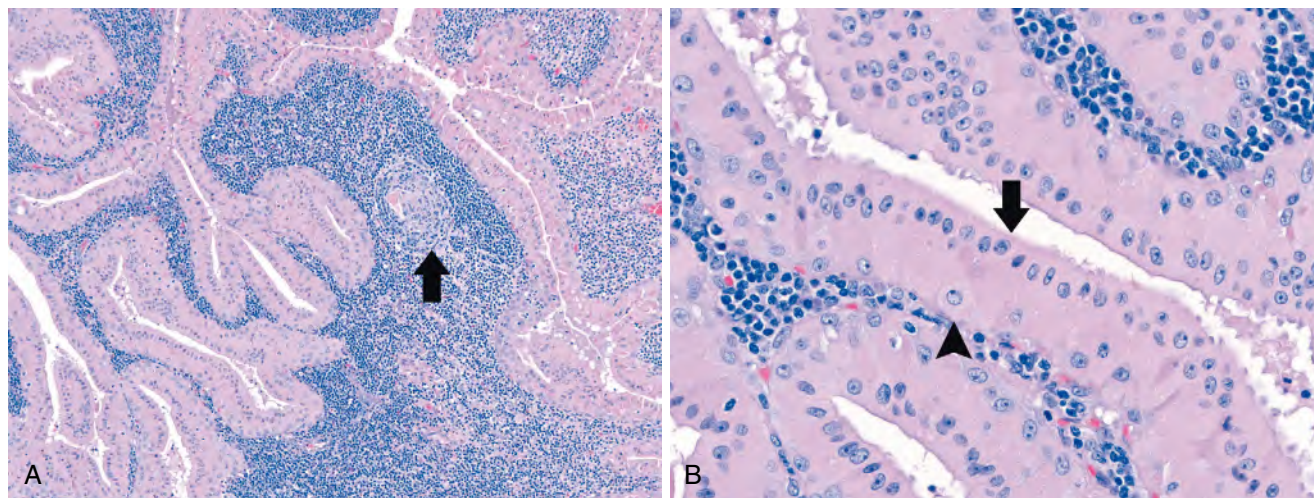


Fig. 16.21 Warthin tumor. (A) Low-power view showing epithelial and lymphoid elements. Note the lymphoid follicular germinal center beneath the epithelium (*arrow*). (B) The epithelium consists of a double layer of oncocytic epithelial cells on a lymphoid stroma, with a columnar luminal layer (*arrow*) associated with a cuboidal layer (*arrowhead*).

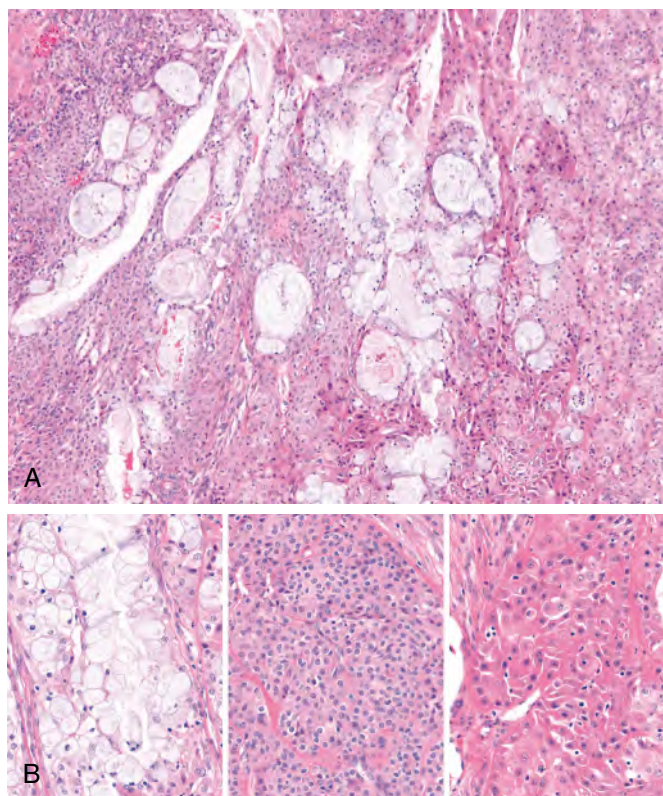


Fig. 16.22 Mucoepidermoid carcinoma. (A) Low-power view showing sheets and microcysts containing cells of varying morphologies. (B) These cells vary from mucinous (*left*), to intermediate (*center*), to squamous (*right*).

The clinical course and prognosis of mucoepidermoid carcinomas reflect the histologic grade. Low-grade tumors may invade locally and recur in about 15% of cases, but they rarely metastasize. The 5-year survival rate is more than 90%. In contrast, high-grade neoplasms are invasive and difficult to excise. Up to 30% recur and 30% metastasize to distant sites, yielding a 5-year survival of only 50%.

Adenoid Cystic Carcinoma

Adenoid cystic carcinomas are relatively uncommon. Around 50% of these tumors occur in the minor salivary glands (particularly the palatine glands). Among major salivary glands, parotid and submandibular glands are most affected. Adenoid cystic carcinomas also occur in the nose, sinuses, upper airways, lung, breast, and other sites. *MYB::NFIB* or *MYBL1::NFIB* gene rearrangements are present in the majority of adenoid cystic carcinomas. Activating *NOTCH1* mutations are present in a subset of adenoid cystic carcinomas with distinctive morphologic appearances and clinical behavior.

MORPHOLOGY

Grossly, adenoid cystic carcinomas are poorly circumscribed, infiltrative, gray-pink lesions. Two variants are observed. *NOTCH1* wild-type tumors are composed of a mixture of basal myoepithelial cells and epithelial cells, which are usually organized in a cribriform growth pattern that resembles a sieve or Swiss cheese. The spaces between the tumor cells are often filled with hyaline material thought to represent excess basement membrane (Fig. 16.23A, left). By contrast, *NOTCH1* mutated tumors are comprised predominantly of epithelial cells and typically grow in solid nests or sheets (Fig. 16.23A, right).

Adenoid cystic carcinomas are unpredictable, slow-growing tumors that tend to invade perineural spaces (see Fig. 16.23B). They regularly recur, and 50% or more eventually recur in distant sites such as the lung, sometimes decades after attempted removal. Thus, although the 5-year survival rate is about 60% to 70%, it drops to about 30% at 10 years and 15% at 15 years. Survival rates are lower in *NOTCH1* mutated tumors, which show an increased proclivity to metastasize to bone and liver. Adenoid cystic carcinomas arising in the minor salivary glands tend to have a poorer prognosis than those arising in the parotid glands.

Acinic Cell Carcinoma

These represent only 2% to 3% of salivary gland tumors and are generally small and well-circumscribed. Most develop in the parotid glands, with the remainder arising in the submandibular glands. Minor salivary glands, which have only a scant number of normal serous acinic cells, are rarely involved. Like Warthin tumors, acinic cell carcinomas can be bilateral or multicentric.

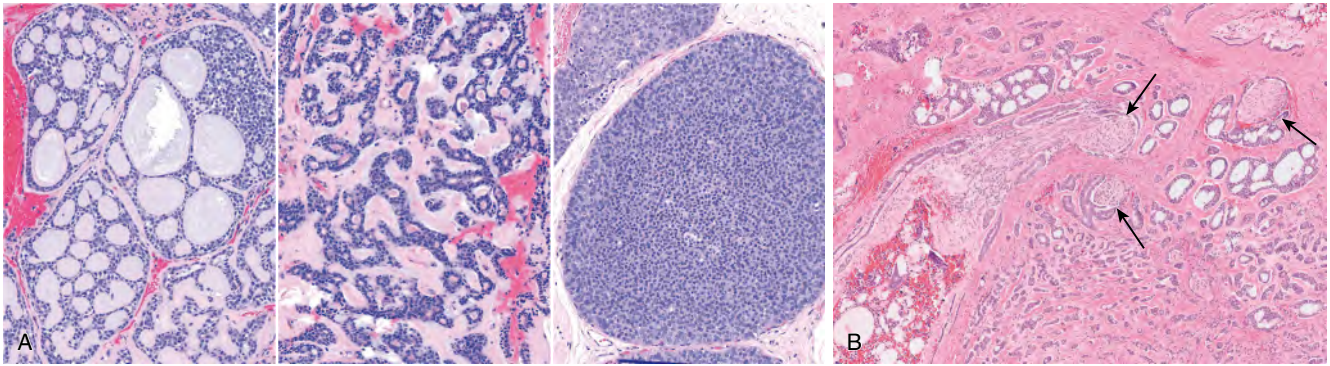


Fig. 16.23 Adenoid cystic carcinoma. (A) Three typical growth patterns including cribriform (left), tubular (center), and solid (right). (B) Perineural invasion (arrows) by adenoid cystic carcinoma.

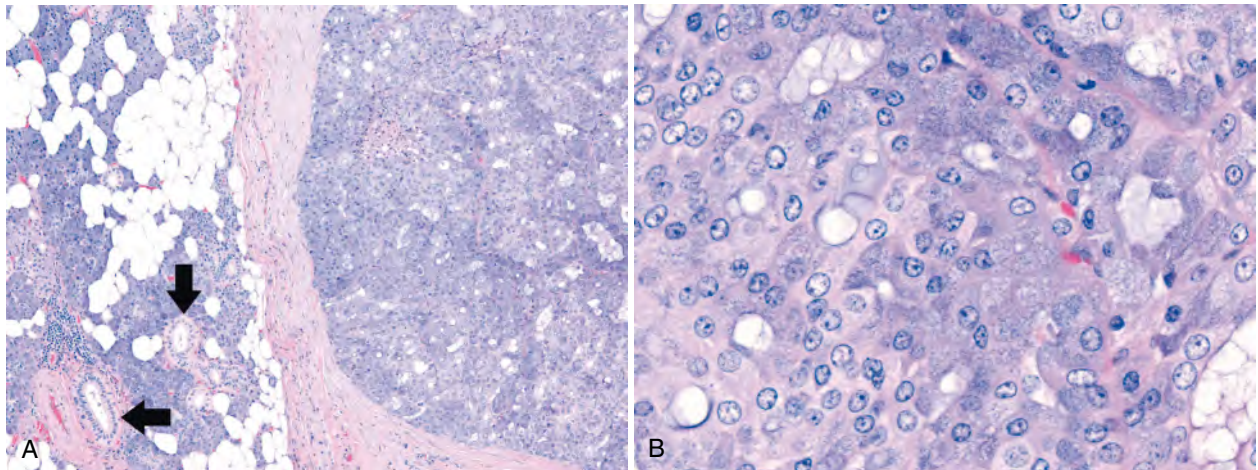


Fig. 16.24 Acinic cell carcinoma. (A) Low-power view showing normal serous parotid tissue with admixed fat and ducts (arrows) on the left and sheets of serous cells in acinic cell carcinoma on the right. (B) High-power image of tumor cells with purple cytoplasmic granules and monotonous round nuclei.

MORPHOLOGY

Histologically, acinic cell carcinomas are composed of cells that resemble normal serous acinar cells of the salivary glands, with small and round nuclei and variable morphology (Fig. 16.24). The cytoplasm typically contains purple granules (**zymogen granules**), but cells can also be clear or vacuolated. The neoplastic cells are organized in sheets, microcysts, glands, follicles, or papillae. Tumors with a sheet-like growth pattern, nuclear pleomorphism, mitoses, and necrosis are referred to as dedifferentiated or having undergone high-grade transformation.

The clinical course of acinic cell carcinomas is reflected by the histologic grade, which considers the degree of nuclear pleomorphism, mitoses, and necrosis. Although recurrence is uncommon after resection, 10% to 15% of tumors metastasize to lymph nodes. The survival rate is approximately 90% at 5 years and 60% at 20 years.

KEY CONCEPTS

Diseases of Salivary Glands

- Sialadenitis is inflammation of the salivary glands and can be caused by trauma, infection, or autoimmune disease.
- Mucoceles are the result of trauma to or blockage of a salivary gland duct that allows saliva leakage into surrounding connective tissue.
- Pleomorphic adenoma, the most common benign salivary gland tumor, is a slow-growing neoplasm composed of a heterogeneous mixture of epithelial and mesenchymal cells.

- Mucoepidermoid carcinoma, the most common malignant salivary gland tumor, shows variable aggressiveness and is composed of a heterogeneous mixture of squamous, intermediate, and mucous cells.

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